

Evaluation and Deployment of 3D-Printed Automatic Weather Stations (3D-PAWS) in Kenya with Integrated Data Visualization and Monitoring Systems

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ABSTRACT

This study presents the deployment and evaluation framework of a 3D-Printed Automatic Weather Station (3D-PAWS) in Kenya. It offers a cost-effective and scalable alternative to conventional Automatic Weather Stations (AWS). Reliable meteorological data is crucial for sectors like agriculture, aviation, and climate monitoring. However, standard AWS systems often face challenges such as high costs, complex maintenance, and limited coverage, particularly in Regional Association I (RA I).

The 3D-PAWS system comprises multiple environmental sensors, including those for temperature, humidity, pressure, rainfall, wind, and solar radiation, all housed within a modular 3D-printed and locally available materials structure. The system runs on a solar-battery setup and uses a cellular-enabled microcontroller to send data in real time to cloud-based platforms. Users can access this data through the CHORDS Portal (<https://3d-fewsnet.icdp.ucar.edu>). We are also developing a custom dashboard for the Kenya Meteorological Department to facilitate ongoing monitoring, visualization, and management of a network of stations.

A significant aspect of the system is its local fabrication approach. 3D printing, assembly, and initial sensor calibration occur in-house at the Kenya Meteorological Service Authority (KMD), Kenya Workshop. This approach improves system adaptability, lowers deployment costs, and promotes sustainable scaling of weather monitoring infrastructure. with collaboration with our development partners CHC, FewsNet, UCAR, UCSB under the EMNP Program.

This paper outlines the system design, deployment architecture, and data infrastructure of 3D-PAWS in Kenya. We propose a comparative analysis with high-end AWS and manual observations to evaluate performance and reliability, as a foundation for further analysis.

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CHAPTER ONE

1 INTRODUCTION

1.1 Background

Reliable and timely weather data is vital for supporting activities like agriculture, aviation, disaster risk management, and climate monitoring. In Kenya, increasing climate variability and the demand for localized forecasting underscore the need for expanding meteorological observation networks. While traditional Automatic Weather Stations (AWS) are highly reliable, they often entail high acquisition costs, complex maintenance, and limited distribution, especially in remote areas.

Recent progress in low-cost sensing technologies, open-source electronics, and 3D printing has led to new weather monitoring solutions. One notable development is the 3D-Printed Automatic Weather Station (3D-PAWS), which provides a flexible, customizable, and affordable method for collecting environmental data. By using 3D-printed parts and commonly available sensors, 3D-PAWS lowers deployment barriers while maintaining acceptable measurement accuracy.

Simultaneously, cloud-based data platforms and visualization tools have changed how meteorological data is accessed and used. Systems like the CHORDS Portal (<https://3d-fewsnet.icdp.ucar.edu>) allow centralized data storage and real-time monitoring. Custom dashboards enhance management capabilities, data visualization, and performance tracking.

1.2 Motivation and Problem Statement

Despite their proven accuracy and reliability, the deployment and expansion of conventional Automatic Weather Stations (AWS) are often constrained by high acquisition costs, specialized maintenance requirements, and dependence on dedicated infrastructure. These factors can limit the density of observation networks, particularly in remote and underserved regions, thereby reducing the availability of localized weather and climate information required for forecasting, disaster risk reduction, agriculture, and climate monitoring.

Recent advances in low-cost sensing technologies and additive manufacturing have enabled the development of affordable weather monitoring solutions capable of complementing conventional observation systems. However, there is limited documentation regarding the large-scale deployment, operational performance, and long-term sustainability of such systems, particularly within Africa. Furthermore, few studies have examined the integration of low-cost weather stations with cloud-based data management platforms and real-time monitoring systems.

There is also a lack of documented operational experiences concerning the maintenance requirements, scalability, and robustness of locally fabricated weather stations operating under diverse environmental conditions. Consequently, the practical viability of these systems as a means of enhancing observation network density remains insufficiently explored.

This study addresses these gaps by investigating the deployment and operational characteristics of the 3D-Printed Automatic Weather Station (3D-PAWS) network in Kenya and examining the associated data infrastructure and visualization systems. In addition, the study provides insights into the scalability, maintainability, and potential contribution of low-cost weather stations towards strengthening meteorological observation networks, particularly in resource-constrained environments.

1.3 Objectives

The main objective of this study is to evaluate and document the deployment of 3D-Printed Automatic Weather Stations (3D-PAWS) in Kenya and their integration with cloud-based visualization and monitoring systems.

The specific objectives are:

- To describe the design and local fabrication of the 3D-PAWS system.
- To document the deployment and expansion of the 3D-PAWS network in Kenya.
- To present the data transmission and visualization infrastructure supporting the network.
- To evaluate the operational performance and reliability of the system.
- To identify challenges encountered during field deployment and maintenance.

1.4 Contributions of the Study

This paper details the design and deployment framework of 3D-PAWS in Kenya, highlighting how it can complement existing meteorological infrastructure. The study focuses on system architecture, local fabrication processes, and data infrastructure, this study evaluates performance against high-end AWS and manual observations in future analyses.

The implementation of 3D-PAWS has progressed from pilot programs to an expanding national network. Initial deployments started with prototype stations through collaborative efforts, leading to large-scale installations across various counties. So far, over seventy stations have been deployed through partnerships with national and international organizations, demonstrating the system's scalability. Field experiences from these deployments have provided valuable information about performance, maintenance needs, and operational challenges in varying environmental conditions.

1.5 Organization of the Paper

The remainder of this paper is organized as follows.

Chapter 2 reviews related work on conventional and low-cost weather monitoring systems.

Chapter 3 presents the design and local fabrication of the 3D-PAWS system.

Chapter 4 describes the national deployment of the network in Kenya.

Chapter 5 discusses the data infrastructure and monitoring systems.

Chapter 6 outlines the methodology adopted for system evaluation,

Chapter 7 presents the performance assessment results.

Chapter 8 discusses operational challenges, lessons learned, discussion & conclusions

Chapter 9 Cost Analysis

Chapter 10 Overall synthesis and achievement of the study objectives

CHAPTER TWO

2 RELATED WORK

2.1 Conventional Automatic Weather Stations

Automatic Weather Stations (AWS) constitute an important component of national, regional and global meteorological observing networks. They provide automated measurements of atmospheric variables and support applications including weather forecasting, climate monitoring, agriculture, hydrology and aviation. Conventional AWS platforms generally integrate multiple meteorological sensors with data acquisition, power, communication and data-management components to provide continuous observations.

High-end AWS systems are designed to provide reliable and accurate measurements over extended operational periods and are commonly deployed where observations require established standards of quality and continuity. Their use within national meteorological services provides an important reference against which alternative and lower-cost observing technologies can be evaluated.

Despite these advantages, conventional AWS installations can involve substantial acquisition, installation and maintenance costs. They may also require specialized technical expertise, dedicated infrastructure, reliable power and communication facilities, and established maintenance and calibration arrangements. These requirements can constrain the rate at which observation networks can be expanded, particularly in locations where resources and infrastructure are limited.

The limitations associated with the cost and logistical requirements of conventional AWS have contributed to increasing interest in alternative observing technologies capable of complementing established meteorological networks.

2.2 Low-Cost and Open-Source Weather Monitoring Systems

Advances in sensor technology, microcontrollers, embedded computing and digital communication have enabled the development of increasingly affordable weather-monitoring systems. Low-cost weather stations commonly combine commercially available sensors with microcontroller-based data acquisition and communication technologies, providing alternatives to more expensive conventional AWS platforms.

Previous applications of low-cost weather-monitoring systems have demonstrated their potential in environmental monitoring, agricultural applications, local weather observation and other situations requiring increased spatial observation density. The use of commercially available components and open-source hardware can also facilitate local assembly, modification and maintenance suitable community customized weather stations

However, affordability does not automatically imply measurement equivalence with conventional reference systems. Low-cost sensors may be affected by calibration requirements, environmental exposure, sensor drift, installation conditions, manufacturing variability and limitations in data quality. Consequently, systematic field validation against established reference observations remains an important requirement when assessing the suitability of low-cost weather-monitoring technologies.

The development of low-cost systems therefore represents not simply a reduction in equipment cost, but an alternative approach to expanding observation capacity through combinations of affordable hardware, local technical capability, digital communication and appropriate quality-control procedures.

2.3 3D-Printed Automatic Weather Stations (3D-PAWS)

2.3.1 Development and Concept of 3D-PAWS

The 3D-Printed Automatic Weather Station (3D-PAWS) represents a development within the broader field of low-cost and replicable meteorological observing systems. The initiative associated with UCAR and FEWS NET under the climate hazard Center (CHC) uses additive manufacturing to support the production of station components while incorporating commercially available sensing and electronic technologies.

The concept was developed around the need for weather-observation systems that could be produced and deployed at comparatively low cost while remaining sufficiently flexible to support different observing environments. The use of 3D-printed components provides opportunities for local production and modification of station structures, while the use of commercially available sensors and electronics supports modularity and replacement.

The 3D-PAWS approach is therefore relevant to meteorological network densification because it combines relatively accessible manufacturing methods with automated observation and digital data transmission.

2.3.2 Modularity and Replicability

An important characteristic of the 3D-PAWS concept is its modular nature. Station components can be produced, assembled, replaced and modified without requiring the entire observing system to be redesigned. This provides potential advantages for deployment in locations where access to specialized meteorological equipment or replacement parts may be limited.

The replicability of the approach also provides opportunities for expanding observation networks through distributed station deployment. Rather than relying exclusively on a small number of expensive conventional stations, locally produced systems can potentially be deployed at additional locations to improve the spatial representation of weather conditions.

Nevertheless, replicability must be accompanied by appropriate calibration, quality assurance and maintenance. The ability to produce many stations does not by itself guarantee consistency between observations, making field validation an important component of any large-scale implementation.

2.3.3 3D-PAWS Applications and Observation-Network Densification

The 3D-PAWS concept has demonstrated potential for deployment in regions where conventional meteorological infrastructure may be difficult or expensive to expand. Its relevance is particularly apparent where there is a need for additional observations at local scales, including agricultural and resource-constrained environments.

For national meteorological services, such systems can potentially function as a complementary observation layer. Conventional AWS can continue to provide established reference observations, while lower-cost systems can increase the number of observation points between conventional stations.

This complementary approach is particularly relevant to Kenya, where differences in topography, climate, land use and rainfall regimes create a need for observations with greater spatial representation.

2.3.4 3D-PAWS and Digital Data Services

The usefulness of an automatic weather station extends beyond the physical measurement of meteorological variables. The ability to transmit, store, visualize and access observations is increasingly important for distributed observing networks.

The 3D-PAWS approach incorporates digital communication and data services to facilitate the movement of observations from individual stations to centralized platforms. This provides a foundation for real-time monitoring, historical data access and management of geographically distributed stations.

The integration of station observations with digital data services is particularly relevant where a network contains numerous remotely located stations. Centralized access can assist with monitoring data continuity, identifying station problems and making observations available for analysis and operational applications.

2.3.5 Relevance of 3D-PAWS to the Study

Although the 3D-PAWS concept provides an established basis for low-cost and replicable meteorological observations, its performance and operational suitability cannot be assumed to be identical across all deployment environments. Differences in climate, elevation, station exposure, maintenance conditions, communications infrastructure and local technical capacity may influence system performance.

This study therefore extends the existing 3D-PAWS concept by focusing on its deployment and evaluation under Kenyan conditions. Particular emphasis is placed on field performance, comparison with Kenya Meteorological Department reference observations, data infrastructure, station monitoring and the potential contribution of the technology to meteorological observation-network densification.

Importantly, the detailed design, sensor configuration, fabrication process, assembly and installation adopted in this study are presented separately in Chapter 3.

2.4 Cloud-Based Meteorological Data Management and Visualization

The increasing deployment of distributed meteorological stations has created a need for effective systems for data transmission, storage, visualization and monitoring. Cloud-based and web-enabled data services provide mechanisms through which observations from geographically distributed stations can be collected and accessed from centralized locations.

CHORDS (Cloud-Hosted Real-time Data Services) provides an example of a platform used to support the collection and management of environmental observations. Such platforms can facilitate access to real-time and historical observations and provide a foundation for managing data from distributed monitoring networks.

In addition to centralized data services, customized dashboards can provide station-level visualization and monitoring capabilities. Such systems can integrate observations with station metadata, geographical information, data-quality indicators and station-health information. This can improve the ability of network managers and technical personnel to identify missing data, communication problems and abnormal observations.

For low-cost observing networks, the integration of hardware with digital monitoring infrastructure is particularly important. A low-cost station that cannot be effectively monitored, maintained and managed remotely may provide limited operational value when deployed across a large geographical area.

This study therefore considers the data-management and visualization component as an integral part of the 3D-PAWS observing system rather than as a separate activity.

2.5 Validation of Low-Cost Meteorological Observations

The deployment of low-cost meteorological sensors requires systematic evaluation against established reference observations. Validation is necessary to determine whether observations reproduce relevant temporal variability and whether differences between the low-cost and reference systems remain within acceptable limits for the intended application.

Common statistical measures used in comparative validation include bias, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), correlation and coefficient of determination (R^2). These measures provide complementary information. Bias indicates systematic differences, MAE describes the average magnitude of absolute errors, RMSE gives greater weight to larger errors, while correlation and R^2 describe the degree to which temporal variability in the reference observations is reproduced.

Data completeness is also important in validation. Statistical results based on a small number of paired observations may not provide the same level of confidence as results derived from longer and more complete records. Consequently, observation coverage and data quality should be considered alongside statistical performance.

For meteorological applications, validation should also consider the characteristics of individual parameters. Temperature, relative humidity, pressure and rainfall are affected by different sensor, exposure and environmental factors. Rainfall, for example, can exhibit substantial spatial variability over relatively short distances, while pressure observations are strongly influenced by elevation and pressure-reduction procedures.

The present study adopts this multi-dimensional validation approach by combining data-quality assessment, statistical comparison, graphical analysis and performance assessment across multiple Kenyan stations.

2.6 Low-Cost Weather Networks and Observation Densification in Africa

Many African countries face challenges associated with sparse meteorological observation networks, high equipment costs, maintenance requirements and limited technical resources a situation that has made Africa be referred as data desert. These challenges can reduce the availability of observations required for weather forecasting, climate services, agricultural applications and local decision-making.

Low-cost and locally maintainable observing technologies provide a potential mechanism for increasing observation density while reducing some of the financial and logistical barriers associated with conventional AWS deployment. However, successful implementation requires more than the availability of inexpensive hardware. Long-term sustainability depends on calibration, maintenance, communications, data management, technical capacity and appropriate integration with existing national meteorological infrastructure.

The African context therefore provides an important setting for evaluating the practical value of technologies such as 3D-PAWS. Evidence from deployment under different environmental conditions can help determine where such systems provide useful complementary observations and where additional technical improvements are required.

For Kenya, the combination of diverse climatic zones, agricultural production areas, arid and semi-arid regions and varying observation density creates a strong context for investigating scalable and locally supportable observing technologies.

2.7 Research Gap

The reviewed literature demonstrates substantial development in conventional AWS technology, low-cost weather-monitoring systems and 3D-PAWS. However, several areas remain insufficiently documented in the context of Kenyan meteorological observation networks.

First, there is limited evidence from multi-station, long-duration field evaluation of 3D-PAWS under diverse Kenyan environmental conditions. Performance demonstrated at an individual site cannot necessarily be generalized to stations operating under different climatic, topographic and environmental conditions.

Second, there is a need for more systematic comparison of 3D-PAWS observations with KMD reference observations across multiple meteorological parameters, including temperature, relative humidity, atmospheric pressure and rainfall.

Third, relatively limited attention has been given to the combined assessment of station performance, data quality, cloud-based data transmission, visualization and operational monitoring within a single 3D-PAWS deployment framework.

Fourth, there is a need to examine the practical implications of locally fabricated stations for meteorological observation-network densification, including operational experience, maintenance requirements and cost considerations.

The present study addresses these gaps through the deployment and evaluation of 3D-PAWS stations across selected Kenyan locations, integration with digital data services and visualization systems, systematic comparison with KMD reference observations, network-level validation and assessment of the implications for observation-network densification.

2.8 Chapter Summary

The chapter has reviewed the development of conventional automatic weather stations, low-cost and open-source weather-monitoring systems, 3D-PAWS, cloud-based meteorological data services, validation approaches and the broader need for observation-network densification in Africa.

The review establishes 3D-PAWS as the principal technological context for the present study while identifying limitations in existing evidence concerning multi-station deployment and validation under Kenyan conditions. It also highlights the importance of integrating physical observation systems with data transmission, visualization, quality control and operational management.

Based on these gaps, the study proceeds in Chapter 3 to describe the system design and local fabrication of the 3D-PAWS stations used in the Kenyan deployment.

CHAPTER THREE

3 SYSTEM DESIGN AND LOCAL FABRICATION

The 3D-Printed Automatic Weather Station (3D-PAWS) is a modular and scalable environmental monitoring system designed to provide meteorological observations using locally fabricated structural components, low-cost sensors, embedded data acquisition and cellular communication. In this study, the fabrication, sensor assembly and calibration/testing, module integration and pre-deployment preparation of the 3D-PAWS stations are undertaken locally at the Kenya Meteorological Department (KMD) workshop. The approach combines additive manufacturing with conventional mechanical assembly, electronic integration and sensor calibration.

The local implementation provides an opportunity to develop practical capacity in 3D printing, station assembly, sensor integration and maintenance while enabling individual components to be produced and replaced locally. The resulting system is designed for deployment in geographically dispersed locations, including areas with limited infrastructure and the arid and semi-arid lands (ASALs) of Kenya, where there is a GSM network.

3.1 Overview of the 3D-PAWS System

The 3D-PAWS system provides observations of key meteorological and environmental variables using a combination of low-cost sensors and a cellular-enabled datalogger. The station measures atmospheric temperature, relative humidity, atmospheric pressure, rainfall, wind speed, wind direction and solar/ultraviolet radiation index. Additional parameters, including heat index, wet-bulb temperature and wet-bulb globe temperature (WBGT), can be derived from the measured observations. Notable feature in 3DPAWS sensors integration is the use of two rain gauges which provide redundancy and quality control aspect in rainfall measurement.

The station is designed as a modular system in which the individual sensing, structural, electronic and power components can be assembled, tested, maintained and replaced independently. This modularity supports local fabrication and facilitates adaptation of the station to different deployment environments.

The completed station incorporates the meteorological sensors on a locally assembled support structure, with the sensing components positioned to provide appropriate exposure to the surrounding environment. The station is connected to a cellular-enabled datalogger that manages data acquisition and transmission to the remote data platform.



Figure 3:13 Completed 3D-PAWS station installed for field operation

3.2 Evolution of the 3D-PAWS Design

The 3D-PAWS system deployed in Kenya has undergone progressive development based on experience gained from prototype construction, field deployment, operation and maintenance. The evolution has involved improvements in the power system, data acquisition and communication, physical configuration, sensor mounting, maintainability and integration with remote data services. For purposes of this study, the development is described in three stages: first-generation, second-generation and the current third-generation configuration.

3.2.1 First-Generation Configuration

The first-generation 3D-PAWS represented the initial implementation of the low-cost automatic weather station concept. The system employed a relatively large battery and solar-power arrangement to support autonomous operation. Data acquisition was performed using a dedicated raspberry datalogger, while communication relied on a SIM-card-based cellular communication system using the available older-generation mobile technology. Although the configuration demonstrated the feasibility of autonomous low-cost weather observations, its relatively large power and communication components increased the physical size and complexity of the installation. Field experience with this configuration provided important lessons concerning power management, communication reliability, equipment protection and maintenance.



Figure 3:2 First-Generation 3D-PAWS

3.2.2 Second-Generation Configuration

The second-generation configuration incorporated improvements arising from experience with the first-generation stations. Particular attention was given to reducing the physical complexity of the installation and improving the arrangement of the meteorological sensors and rainfall gauges. The sensors were arranged around a central support structure, with the rainfall gauges positioned separately from the main sensor assembly. The electronic components were housed in a protective enclosure, while the overall station structure was adapted for improved field deployment and maintenance.

This generation provided additional operational experience concerning sensor exposure, Data acquisition structural stability, environmental protection, rainfall measurement, cellular connectivity and accessibility of components during maintenance.



Figure 3:3 Second -Generation 3D-PAWS

3.2.3 Current Third-Generation Configuration

The current configuration represents the latest stage in the development of the 3D-PAWS system used in this study. The design incorporates lessons obtained from previous generations and focuses on a compact, modular and field-maintainable station architecture.

As shown in Figure 3.4, the current station incorporates a central support structure carrying the upper meteorological sensors, a radiation shield for the temperature and humidity sensing assembly, a protected electronics enclosure, and separately positioned rainfall gauges.

The current generation is also integrated using modern particle Boron datalogger, data-management architecture in which observations are transmitted through cellular communication to the CHORDS platform for centralized storage, visualization using Grafana and historical retrieval. The system will further be complemented by a custom monitoring dashboard which is being developed.

The progression from the first to the current generation therefore reflects a transition from an initial not only autonomous station concept toward a more compact, maintainable, connected and remotely monitored observation system.

The particle Boron datalogger provides for seamless integration with a wide variety of systems as well as configuration of a local sim card capabilities.



Figure 3:4 Current Third-Generation 3D-PAWS

Table 3:1 3D-PAWS Generation Major Comparison

Generation	Major characteristics	Development emphasis
1st Generation	Large battery and solar system; older SIM-based communication and raspberry datalogger	Initial proof of autonomous operation
2nd Generation	Improved sensor arrangement and physical station configuration. Operates on Particle Boron datalogger	Field deployment and maintainability
3rd Generation Current	Compact integrated structure; improved sensor mounting; protected electronics; cellular/cloud connectivity; dashboard integration. Operates on Particle Boron datalogger	Scalability, maintainability and network monitoring

3.3 System Design and Architecture

The 3D-PAWS system comprises five principal subsystems:

1. Mechanical and structural subsystem: consisting primarily of 3D-printed components and supporting mechanical elements;
2. Meteorological sensing subsystem: comprising sensors for temperature, humidity, pressure, rainfall, wind and radiation;
3. Data acquisition subsystem: based on a Particle Boron datalogger;
4. Power subsystem: comprising solar power generation and battery storage.
5. Communication and data-management subsystem: providing cellular transmission to the CHORDS platform and associated visualization and analysis systems.

The overall system therefore combines locally fabricated mechanical components, locally sourced materials and commercially available sensing and electronic components. This approach allows the structural components to be produced locally while retaining the use of pre-calibrated sensing elements and established communication technologies.



Time
Health
Battery Voltage
Battery charge status
Cell signal strength
Rain Gauge 1
Rain Gauge 2
Rain Gauge 1 Total Today
Rain Gauge 2 Total Today
Rain Gauge 1 Total Prior
Rain Gauge 2 Total Prior
BMX Temperature 1
BMX Pressure 1
MCP Temperature 1
SHT Temperature
SHT Humidity
SI1145 Visible 1
SI1145 Infrared 1
SI1145 Ultraviolet 1
Wind Speed
Wind Direction
Wind Gust
Wind Gust Direction
Heat Index
Wet Bulb Temperature
Wet Bulb Globe Temperature

Figure 3:5 3D-PAWS system architecture

3.4 Training and Capacity Development

The local fabrication programme was supported by practical training of the participating technical team in the use of 3D-printing technology and the assembly of 3D-PAWS components. The training provided participants with practical exposure to the fabrication workflow, including the handling of printed components, assembly of sensors, station structures and integration of the sensing and electronic components.

The training was important because the adoption of 3D-PAWS involved not only the deployment of a new type of weather station but also the development of local technical capability to fabricate, assemble, maintain and reproduce the stations.

The training and subsequent practical work enabled the team to gain hands-on experience in additive manufacturing and station assembly. This local capacity provides an important foundation for continued maintenance and expansion of the 3D-PAWS network.



Figure 3:6 Technical team participating in 3D-PAWS fabrication and assembly training at the KMD workshop

3.5 Local Fabrication at the KMD Workshop

A key component of the implementation is the establishment and use of a local fabrication and assembly workflow within the KMD workshop. Rather than treating the station as a complete externally manufactured unit, the structural components are produced and assembled locally, while the sensors and electronic components are integrated into the locally fabricated structure.

The workshop environment provides space for 3D printing, component storage, mechanical assembly, wiring, soldering, sensor integration and testing. The photographs from the workshop show multiple batches of printed components, including radiation shields, funnels, sensor housings, mounting components and other structural parts prepared for station assembly.

The availability of multiple printed components at different stages of preparation demonstrates the modular nature of the fabrication process. Components can be produced in batches and organized for subsequent station assembly. This approach is particularly useful for network expansion because the same structural components could be reproduced as additional stations are prepared.



Figure 3:7 Locally fabricated 3D-PAWS components at the KMD workshop.

3.5.1 3D Printing and Production of Structural Components

The structural components of the stations are produced using desktop 3D printers available at the KMD workshop. The printing process involves preparing the required digital component designs, processing them for additive manufacturing and producing the components individually according to their intended function.

The workshop photographs show the production of several types of components, including conical structures, radiation-shield components, sensor housings, mounting brackets, rain-gauge components and other mechanical parts.



Figure 3:8 Desktop 3D printers used for local production of 3D-PAWS components at the KMD workshop.



Figure 3:9 Examples of 3D-printed structural and sensor-mounting components.

The use of additive manufacturing enables components to be produced individually and in batches. This is particularly advantageous for components that may require replacement during maintenance because the structural parts can be reproduced locally rather than requiring replacement of an entire station this being the main leverage of the 3DPAWS station.

3.5.2 Component Inspection and Preparation

Following printing, the components are inspected and prepared for assembly. Inspection involves checking the printed parts for visible defects, dimensional conformity and suitability for integration with the corresponding mechanical and sensing components.

The printed components are organized according to their intended application before assembly. This includes the preparation of sensor housings, radiation shields, rainfall components, wind-measurement components and structural supports.



Figure 3:10 Printed components prepared for assembly, integration and installation

The modular nature of the components allows individual parts to be inspected and, where necessary, reproduced before the station is assembled. This provides a practical quality-control step within the local fabrication workflow.

3.5.3 Mechanical Assembly and Sensor Integration

Following inspection, the printed components are assembled to form the physical station structure. Sensor integration is subsequently undertaken by mounting the individual sensors and connecting them to the datalogger.

The assembly process includes:

- fitting the structural supports;
- mounting the temperature, humidity and pressure sensing assembly which runs on the same data bus;
- assembling the tipping-bucket rain gauges;
- assembling the cup anemometers and integrating the wind-vane mechanism;
- installing radiation-sensing components;
- fitting the protective enclosure;
- routing and securing electrical connections; and
- connecting the sensors to the Particle Boron datalogger.

The rain gauge and wind direction sensors are soldered and assembled using Hall-effect components and integrated locally as part of the assembly process.



Figure 3:11 Sensor components and electrical connections during 3D-PAWS assembly.

The photographs demonstrate that assembly involves both mechanical and electronic integration rather than simply mounting preassembled sensors. This local integration allows the station configuration to be inspected and tested before field deployment.

3.5.4 Local Fabrication and Assembly Workflow

The overall fabrication process follows a sequential workflow:

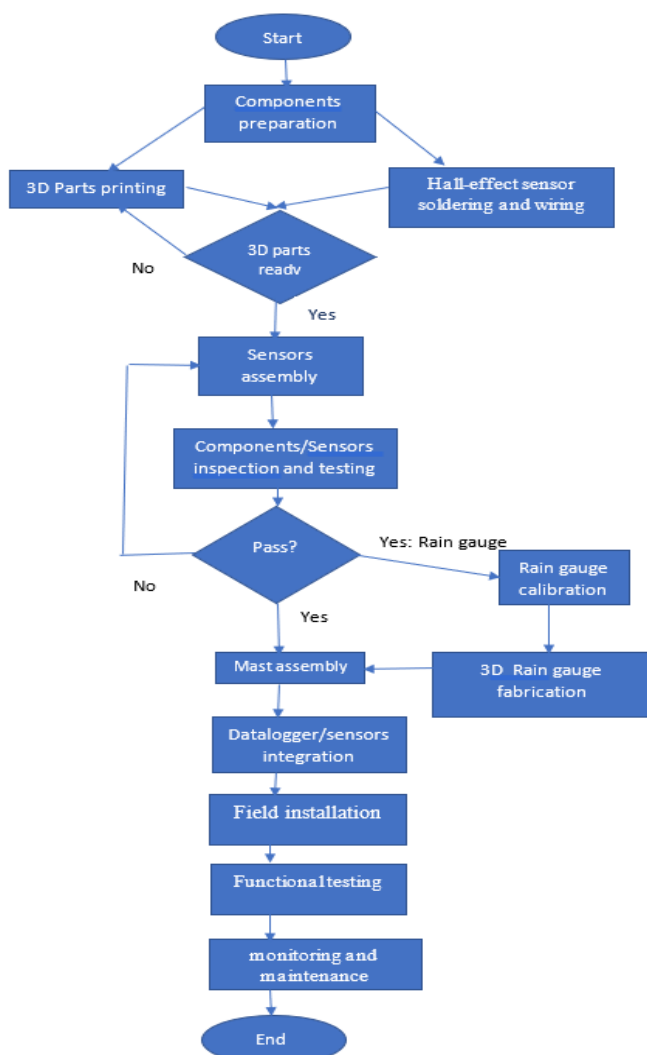


Figure 3:12 Local 3D-PAWS fabrication, assembly, calibration and deployment workflow

The above workflow demonstrates the integration of additive manufacturing, electronics, meteorological instrumentation and field deployment within a single local production process.

3.6 Sensor Suite

The 3D-PAWS system incorporates a range of sensors for measuring the primary meteorological parameters required for observation and validation.

Table 3:2 Primary meteorological sensors used in the 3D-PAWS system

Meteorological Parameter	Sensor/Component	Measurement Principle	Unit
Air Temperature	MCP9808	Digital temperature sensing	°C
Relative Humidity	SHT31	Capacitive humidity sensing	%
Atmospheric Pressure	BMP280/BME280	MEMS barometric pressure sensing	hPa
Wind Speed	Cup anemometer + Hall-effect sensor	Pulse/event detection	m s ⁻¹
Wind Direction	AS5600-based wind vane	Magnetic rotary position sensing	°
Rainfall	Tipping-bucket rain gauge + Hall-effect sensor	Bucket-tip event detection	mm
Solar Radiation	SI1145	Photodiode-based sensing	Counts
Ultraviolet Radiation	SI1145	UV sensing	Index

The temperature, pressure, relative-humidity sensors and solar radiation are factory-calibrated sensing devices. During station fabrication, these sensors are integrated into the locally assembled structure and subsequently subjected to functional checks.

The rainfall and wind systems require additional mechanical and electronic assembly and integration because their measurements depend on physical movement and event detection. Hall-effect sensors are therefore incorporated into the tipping-bucket rain gauge and cup anemometer assemblies.

3.6.1 Derived and Station-Health Parameters

In addition to the primary meteorological observations, the system provides derived and station-health parameters that support operational monitoring and subsequent analysis.

Table 3:3 Derived and station-health parameters

Parameter	Source/Method	Unit
Heat Index	Derived from temperature and relative humidity	°C
Wet-Bulb Temperature	Derived psychometrically	°C
Wet-Bulb Globe Temperature (WBGT)	Derived parameter	°C
Battery Voltage	Internal monitoring circuit	V
Battery Charge Status	Power-system monitoring	C
Cellular Signal Strength	Boron cellular modem	dBm
Station Health	System status monitoring	H

These additional parameters provide information not only about the meteorological environment but also about the operational condition of the station. Battery status and cellular signal strength, for example, can be used to assess the ability of a remote station to continue acquiring and transmitting observations.

3.6.2 Sensor Calibration and Pre-Deployment Testing

Prior to field deployment, the sensing components and completed stations undergo assembly, calibration and functional testing. The MCP9808 temperature sensors, SHT31 relative-humidity sensors and BMP280/BME280 pressure sensors are factory-calibrated. These sensors are integrated into the station during

the local assembly process and subjected to functional checks to confirm appropriate operation and communication with the datalogger.

Additional calibration procedures are applied to measurement components whose operation depends on mechanical movement and event detection.

a) Rain-gauge calibration

The tipping-bucket rain gauges are calibrated using the tipping method. The procedure is used to verify the relationship between individual bucket-tip events and the corresponding rainfall amount recorded by the system. The Hall-effect sensor is integrated into the tipping mechanism to detect each bucket-tip event electronically. The resulting pulses are received by the datalogger and converted into rainfall observations.

b) Wind-speed sensing

The cup anemometers are assembled with Hall-effect sensing components to detect rotational movement and generate measurement events corresponding to wind-speed observations. The sensor connections are soldered and tested during the assembly process.

c) Wind-direction calibration

The wind-vane system is calibrated to ensure that the electronic direction output corresponded correctly to the physical orientation of the vane. The AS5600-based sensing arrangement provides the angular position required to derive wind direction.

The calibration of the sensors and testing is done using a Raspberry pi Program



Figure 3:13 3D-PAWS measurement components during assembly and calibration.

Following sensor-level testing, the complete station is subjected to functional checks covering:

- sensor response;
- datalogger operation;
- electrical connections;
- power supply & battery operation;
- cellular communication;
- data transmission; and
- consistency of observations received by the data platform.

This combined sensor-level and system-level testing provided a final check before field installation.

3.7 Datalogger Architecture

All sensors are interfaced with a Particle Boron microcontroller configured as the station datalogger. The Boron performs the primary data acquisition, processing and communication functions.

The datalogger receives measurements from the connected sensors, processes the observations according to the station configuration and transmits the resulting data through its cellular communication interface.

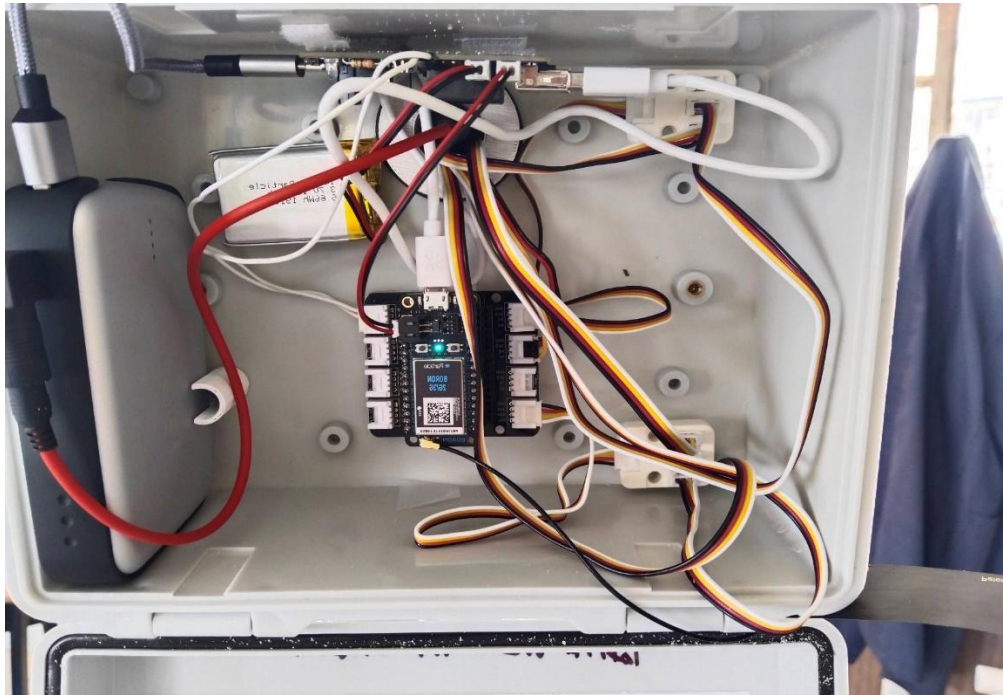


Figure 3:14 Particle Boron datalogger used for 3D-PAWS data acquisition and communication.

The use of an integrated cellular-enabled datalogger reduces the need for separate communication hardware and allows the station to operate as a remotely connected observation platform.

The data flow can therefore be summarized as:

Sensor measurements → Boron data acquisition → data processing → cellular transmission → CHORDS → visualization and analysis.

3.7.1 Power Supply and Communication System

The station is powered by a combination of solar energy and battery storage, enabling autonomous operation at remote and off-grid locations. The Particle Boron datalogger provides cellular communication capability and is configured with a SIM card for transmission of observations to the CHORDS platform.

The communication configuration also provides flexibility for local deployment. The Boron datalogger can be configured to operate using a locally available cellular-network SIM card, allowing the most suitable mobile network to be selected according to network availability at the deployment site.

This arrangement enables observations to be transmitted from geographically distributed stations without requiring fixed internet infrastructure at each observation site. The combination of solar power, battery storage and cellular communication therefore supports autonomous operation and remote monitoring of the deployed stations.

3.8 Station Assembly and Field Preparation

After completion of fabrication, sensor integration and testing, the stations are prepared for field installation. The completed station consisted of the locally fabricated structural components, integrated meteorological sensors, Particle Boron datalogger, power system and protective enclosure.



Figure 3:15 Fully assembled 3D-PAWS station prior to field deployment.

The final assembly is designed to allow the individual sensing components to remain accessible for maintenance and replacement while providing adequate protection for the electronic components. The modular construction also allows stations to be prepared and tested before transportation to their respective deployment locations.

3.8.1 Field Deployment of 3D-PAWS

Following fabrication, sensor integration, calibration, and functional testing, the 3D-PAWS stations were deployed to selected locations within the observation network. Field deployment involved transportation of the assembled stations to the selected sites, physical installation, sensor positioning, power-system setup, communication configuration, and verification of data transmission.

The deployment process was undertaken with consideration of the geographical and environmental characteristics of each site. Station coordinates and elevation were recorded to support station identification, network mapping, data analysis, and comparison with reference observations.

The principal activities undertaken during field deployment included:

1. Site identification and assessment;
2. Transportation of station components and equipment;
3. Physical installation of the station structure;
4. Installation and positioning of meteorological sensors;
5. Installation of the rainfall measurement system;
6. Power-system installation and verification;
7. Configuration and integration of the Particle Boron datalogger;
8. Cellular communication configuration and Verification of sensor measurements
9. Verification of data transmission to CHORDS; and
10. Documentation of station coordinates, elevation, configuration, and operational status.

After installation, the stations were observed to confirm that measurements were being acquired correctly and transmitted to the cloud-based data platform. Where necessary, configuration and physical adjustments are undertaken before the station is considered operational.

The field deployment process followed a sequential workflow:

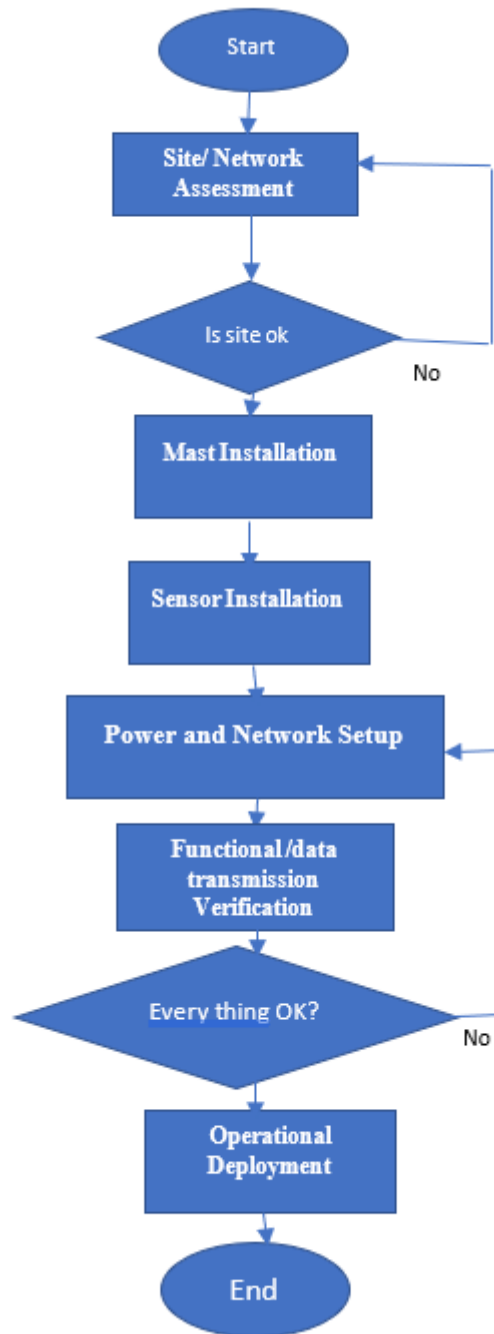


Figure 3:16 Field Deployment Workflow

This workflow is intended to minimize installation errors and ensure that the station is capable of producing and transmitting observations before being incorporated into the observation network.

3.8.2 Challenges Encountered During Field Deployment and Maintenance

Field deployment and subsequent maintenance of the 3D-PAWS stations presents several practical and technical challenges. These challenges are documented as part of the implementation process because they provide important information for assessing the practicality and scalability of locally fabricated 3D automatic weather stations.

a) Site Accessibility and Logistics

Some deployment sites present logistical challenges associated with transportation, accessibility, and the movement of equipment and personnel. The geographical distribution of the stations requires coordination of field activities and, in some cases, additional planning for access to remote locations.

These factors affect the time required for installation, inspection, troubleshooting, and maintenance.

b) Environmental Exposure

The stations are continuously exposed to outdoor environmental conditions. Exposure to rainfall, dust, solar radiation, wind, insects and other environmental factors can affect the physical condition and performance of station components.

Consequently, field maintenance is required, periodic inspection of sensor mounting arrangements, protective enclosures, cables, connectors, and other exposed components.

c) Cellular Network Connectivity

Reliable cellular connectivity is necessary for transmission of observations from the stations to the CHORDS platform. Variations in cellular signal strength or temporary communication failures could result in delayed or missing observations.

This highlights the importance of monitoring communication status and signal strength as part of station health monitoring.

d) Sensor and Component Reliability

The operation of multiple low-cost sensors within an outdoor environment introduces challenges related to sensor stability, calibration, physical protection, and long-term reliability. Sensor performance therefore require periodic verification against expected observations and, where necessary, adjustment or replacement.

e) Rainfall Measurement

Rainfall measurement presents additional operational considerations because the tipping-bucket rain gauge requires appropriate installation, levelling, cleanliness, and protection from physical obstruction. Accumulation of debris or other physical interference can affect rainfall measurements and therefore requires periodic inspection.

f) Data Transmission and Data Gaps

Temporary interruptions in communication, power, or station operation could result in periods of missing observations. These gaps require investigation to determine whether they resulted from sensor issues, power problems, communication failures, or other station-level conditions.

The experience reinforced the importance of combining station health monitoring with meteorological data-quality assessment.

g) Maintenance in a Distributed Network

As the number of deployed stations increases, conducting physical maintenance at every station becomes progressively more resource-intensive. The geographical distribution of stations therefore creates a need for remote monitoring capabilities that can help identify stations requiring physical intervention before field teams are dispatched.

This consideration informs the development of the custom dashboard, which is designed to provide station-level information such as communication status, battery condition, signal strength, sensor availability, and last-seen status.

3.9 Lessons from Field Deployment

The field deployment experience provides practical information that informs subsequent refinement of the 3D-PAWS system. In particular, the experience demonstrate that successful deployment requires more than sensor assembly and calibration. Site selection, physical installation, power management, communication availability, environmental protection, maintenance procedures, and remote monitoring are all important components of operational deployment.

The challenges encountered during deployment and maintenance are therefore incorporated into subsequent system development and monitoring approaches. The development of the CHORDS-based data infrastructure, Grafana visualization, and custom monitoring dashboard provides mechanisms for improving remote visibility of station performance and reducing the need for unnecessary physical site visits.

3.9.1 Maintenance and Component Replacement

The modular construction of the 3D-PAWS system facilitates maintenance and replacement of individual components. Structural components that become damaged can be reproduced locally using the 3D-printing facilities, while individual sensors, electronic components and mechanical assemblies can be replaced without requiring replacement of the complete station.

This capability is particularly relevant to a distributed observation network in which stations may be deployed across locations with different environmental conditions and varying levels of accessibility. Local fabrication therefore provides an avenue for reducing the logistical dependence associated with sourcing complete replacement stations or specialized structural components.

The approach also provides a basis for future modification and upgrading of the stations as improved sensors, communication technologies or monitoring requirements become available.

3.10 Chapter Summary

The chapter presented the development of the 3D-PAWS system, covering its evolution, local fabrication, sensor integration, testing, field deployment, and maintenance. The progression from the earlier generations to the current configuration reflects improvements in station design, power systems, data acquisition, communication, and maintainability. The chapter also identified practical challenges encountered during field deployment and maintenance, providing important lessons for further network expansion. The resulting system provides the technical foundation for the data transmission, monitoring, and validation activities presented in the subsequent chapters.

CHAPTER FOUR

4 NATIONAL DEPLOYMENT OF 3D-PAWS IN KENYA

4.1 Evolution of the Network

The implementation of the 3D-Printed Automatic Weather Station (3D-PAWS) network in Kenya has progressed from an initial pilot phase of 25 stations to an expanding operational network of 59 stations. The initial deployments are undertaken through collaborative efforts involving the Kenya Meteorological Service Authority (KMD), the Famine Early Warning Systems Network (FEWS NET), and the University Corporation for Atmospheric Research (UCAR). These collaborations provided the foundation for testing, refining and progressively expanding the 3D-PAWS approach under Kenyan environmental and operational conditions.

The expansion of the network has demonstrated the potential of locally fabricated, low-cost automatic weather stations to complement existing conventional meteorological observation infrastructure. The modular design of the 3D-PAWS platform, together with the ability to locally fabricate structural components and replace individual parts, has supported relatively rapid deployment and simplified maintenance.

The deployment has progressively extended the observation network across different geographical and climatic environments in Kenya. This includes areas with varying elevations, climatic conditions and levels of existing meteorological observation coverage. The expansion therefore provides an opportunity to increase observation density and improve the availability of near-real-time meteorological observations, particularly in locations that are inadequately served by conventional observation infrastructure.

4.2 3D-PAWS and the National Synoptic Observation Network

The national deployment of 3D-PAWS was undertaken within the broader context of Kenya's existing meteorological observation network. The conventional synoptic network provides an established framework for systematic meteorological observations and serves as an important source of data for weather monitoring, forecasting and climate applications.

Rather than replacing conventional synoptic and automatic weather stations, the 3D-PAWS network provides an opportunity to complement and densify the existing observation network. The relatively low cost, modular construction and flexible deployment characteristics of 3D-PAWS make it possible to establish additional observation points in areas where the density of conventional stations is limited.

The distribution of the 3D-PAWS stations was therefore considered in relation to existing meteorological observation infrastructure, with emphasis on identifying opportunities to improve spatial coverage. Figure 2 presents the distribution of the deployed 3D-PAWS stations in relation to the existing meteorological observation network.

The combined network provides a basis for increasing the spatial availability of meteorological observations while retaining the established national observation infrastructure as an important reference and operational component.

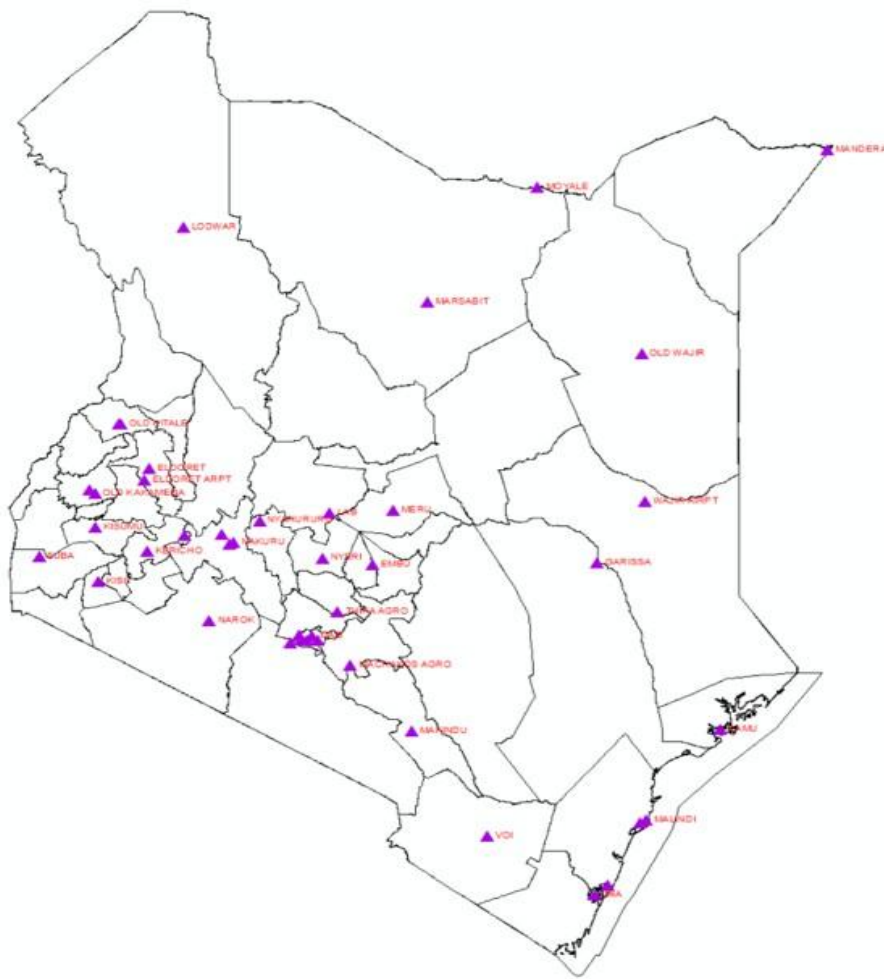


Figure 4:1 Distribution of Synoptic Manual stations

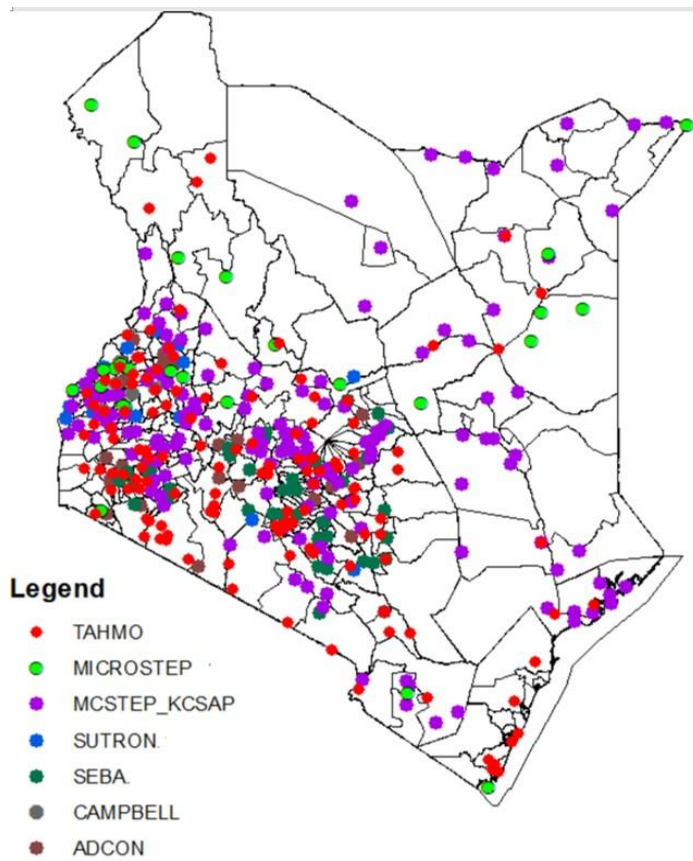


Figure 4:2 Distribution of other Automatic Weather Stations

4.3 Station Selection and Deployment

4.3.1 Station location Selection

Selection of deployment locations is an important component of expanding the 3D-PAWS observation network. Locations are considered with the objective of increasing observation coverage while ensuring that the stations could be practically installed, operated and maintained.

The station-selection process considered the geographical distribution of existing observation stations and the need to improve coverage in areas with limited observations. Consideration was also given to the environmental and operational characteristics of potential sites, including geographical location, elevation, local exposure, accessibility, availability of cellular communication and suitability for installation of the station and its supporting infrastructure.

Where possible, deployment locations are selected to provide observations representative of the surrounding environment and to support applications requiring localized meteorological information. Particular attention was given to areas where increased observation density could support weather monitoring, climate analysis, agricultural applications and early-warning activities.

The use of a modular and relatively low-cost platform provided flexibility in selecting additional sites as deployment requirements evolved. This allows the network to progressively expand beyond the initial pilot locations and respond to identified observation gaps.

Some stations were installed in KMD Enclosures parallel to conventional AWS and manual stations observatories for purposes of data comparisons.

4.3.2 Installation and Site Preparation

Following site selection, the stations are assembled, tested and installed at the designated locations. Installation involved positioning the station structure, mounting the meteorological sensors, connecting the sensor wiring to the datalogger, configuring the power supply and establishing cellular communication.

The station installation process included verification of the physical orientation and positioning of the sensors to ensure that the observations are not unnecessarily affected by nearby structures or other obstructions. The wind-measurement components are positioned to allow representative exposure, while the temperature and humidity sensing system was installed within the radiation-shield assembly.

The tipping-bucket rain gauge was installed in an appropriate exposed position to minimize interference from surrounding structures. The rain-gauge system was also checked to ensure that it was correctly positioned and functioning before the station was commissioned.

Following physical installation, the datalogger was configured for communication using the available cellular network. A local SIM card can also be used in the Particle Boron datalogger, allowing the station to operate using a suitable mobile network available at the deployment location. This provides flexibility where cellular-network availability differs between sites and avoids dependence on fixed internet connectivity.

The installed station is subsequently powered and monitored to confirm sensor operation, data acquisition and successful transmission of observations to the CHORDS platform.

4.3.3 Commissioning and Verification

After installation, each station Undergo commissioning and operational verification before being considered part of the observation network. The verification process includes checking the operation of the individual sensors, power supply, datalogger, cellular communication and data transmission.

The observations transmitted by the station are checked through the CHORDS platform to confirm that the station is communicating and that the expected meteorological parameters were being received. Particular attention was given to temperature, relative humidity, atmospheric pressure, rainfall, wind speed and wind direction, together with system-health parameters such as battery voltage and cellular signal strength.

Where problems are identified during commissioning, corrective actions are undertaken before the station is placed into operational use. The modular construction of the system facilitates replacement, adjustment or repair of individual components without requiring replacement of the complete station.

4.4 Spatial Distribution of Stations

The deployed stations are distributed across various regions of Kenya, covering coastal areas, highlands, agricultural zones, and arid and semi-arid lands (ASALs). The broad geographical coverage exposes the stations to different climatic conditions and provides an opportunity to assess the robustness of the 3D-PAWS system under varying environmental conditions.

The deployment of 3D-PAWS has resulted in stations being distributed across a range of geographical and environmental settings in Kenya. The network includes stations located in different counties and at varying elevations, providing an opportunity to evaluate the performance of the system under diverse environmental conditions.

The geographical distribution is important because Kenya experiences substantial spatial variability in climate, topography and weather conditions. A denser observation network can therefore provide additional localized observations that may complement observations from the existing national meteorological network.

Table 4:1 Distribution of 3D-PAWS Stations by Major Climatic Regions

Climatic Region	Representative Locations
Coastal Region	Mombasa, Malindi, Kilifi, Lamu
Highlands	Nyeri, Nyahururu, Eldoret, Kericho
Agricultural Zones	Kitale, Embu, Meru, Kakamega
Forest Ecosystems	Kakamega Forest, Mau Region
ASAL Regions	Turkana, Marsabit, Wajir, Garissa, Mandera

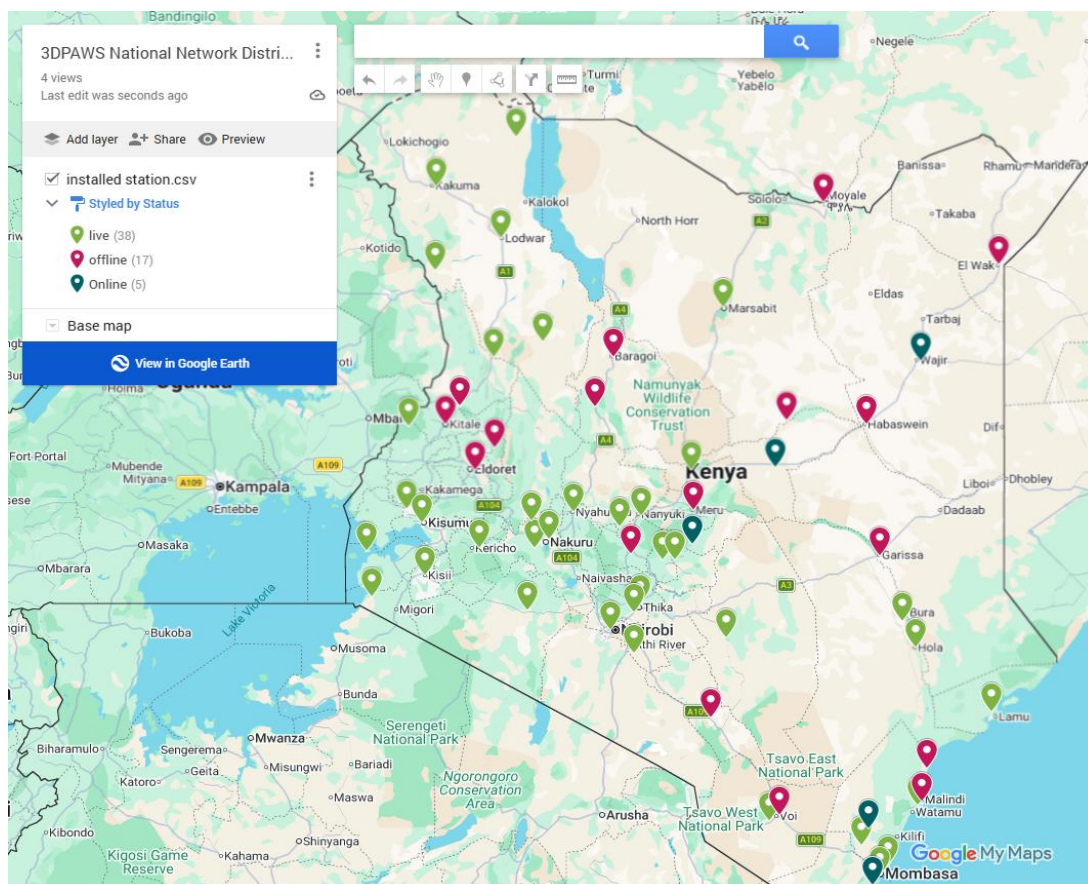


Figure 4:3 Spatial distribution of 3D-PAWS stations in Kenya.

4.4.1 Station Inventory

A total of **59 3D-PAWS stations** have been deployed across Kenya. The stations are distributed across different counties, geographical regions, climatic environments, and host institutions.

The inventory below provides the geographical characteristics of each station, including the station name, latitude, longitude, and elevation. This information was used to characterize the spatial distribution of the network and assess the range of environmental conditions represented by the deployed stations.

4.4.2 Elevation Characteristics

The deployed 3D-PAWS stations span a substantial range of elevations, from near sea level in the coastal region to more than 2,700 m above mean sea level in the highland regions. This elevation range provides an opportunity to assess the operation of the stations under diverse topographic and climatic conditions.

The lowest-elevation station in the current inventory is the Kenya Ports Authority–Mombasa station, located at an elevation of approximately 3 m above mean sea level. The highest-elevation station is the Kenya Agricultural and Livestock Research Organization (K

Table 4:2 Lowest & Highest Elevations

	Station Name	Latitude	Longitude	Elevation(M)
Lowest Elevation	Kenya Ports Authority- Mombasa	-4.058333	39.647778	3
Highest Elevation	Kenya Agricultural and Livestock Research Organization (Kalro) Nyahururu	0.03813	36.36339	2732

4.4.3 Climatic Zones Covered

The geographical distribution of the 3D-PAWS network enables observations across several environmental and climatic settings in Kenya. The deployed stations include locations within:

- Coastal regions;
- Highland regions;
- Agricultural zones;
- Arid and semi-arid lands (ASALs); and
- Forest ecosystems.

The representation of different climatic and environmental conditions provides a basis for assessing the adaptability and operational performance of the 3D-PAWS system under varying environmental conditions.

It also increases the potential application of the observations for agricultural monitoring, climate services, early warning, environmental monitoring, and other applications requiring higher spatial density of meteorological observations.

4.4.4 Host Institutions and Partnerships

The national deployment of the 3D-PAWS network has been supported through collaboration among KMD, FEWS NET, UCAR, research institutions, educational institutions, airports, and other local stakeholders. These partnerships have provided suitable sites for station installation and contributed to the expansion and operational sustainability of the network.

Examples of host institutions and facilities include:

- Kenya Airports Authority facilities;
- Kenya Agricultural and Livestock Research Organization (KALRO);
- Schools and universities;
- Community and other local institutions.

The involvement of host institutions is particularly important for maintaining stations after installation. Providing secure sites and facilitating access for routine inspection and maintenance contributes to the continued availability of observations and reduces the logistical challenges associated with operating a geographically distributed network.

4.5 Upgrade and Expansion of the Network

Following the initial deployment of the 25 stations, several stations underwent subsequent upgrades to improve their measurement capability and operational performance. Early installations were enhanced through the addition of wind and radiation sensors, thereby expanding the range of meteorological variables observed by the stations. Maintenance activities were also undertaken as required and included inspection, troubleshooting, repair and replacement of faulty or damaged components.

The modular design of the 3D-PAWS system enabled these upgrades and maintenance activities to be undertaken without requiring complete replacement of existing stations. Individual components could be added, replaced or modified while retaining the main station structure and datalogger architecture. This provided a practical approach for progressively improving the capability of stations as operational requirements evolved.

The ability to locally fabricate structural components and assemble, modify and maintain the stations has supported the progressive expansion of the network. Rather than requiring complete replacement when

individual components fail or when additional measurement capabilities are required, the modular architecture allows stations to be upgraded incrementally. This characteristic is particularly relevant to long-term deployment, where maintenance, component replacement and technological improvements are expected over the operational life of the observation network.

The progression from 25 pilot stations to 59 deployed stations demonstrates the scalability of the 3D-PAWS approach and its potential to contribute to increased spatial density of meteorological observations in Kenya. The network has expanded across different geographical and environmental settings, providing additional observation points that complement the existing meteorological observation infrastructure.

Continued expansion of the network can be guided by identified observation gaps, application requirements, network performance, availability of suitable installation sites and communication infrastructure. The ability to configure the Particle Boron datalogger with a locally available cellular SIM card further supports deployment across geographically dispersed locations where mobile-network availability may vary.

The growing network also provides an expanding observational dataset for evaluating 3D-PAWS performance against established meteorological observations. The resulting observations provide the basis for the comparative analysis, validation and statistical assessment presented in the subsequent chapters.

Table 4:3 3D-PAWS Installation Summary

Parameter	Value
Number of stations deployed	59
Communication system	Cellular
Power supply	Solar and battery
Minimum elevation	Approximately 3 m above mean sea level
Maximum elevation	Approximately 2,340 m above mean sea level
Mean station elevation	Approximately 1,067 m
Climatic zones covered	Coastal, Highlands, Agricultural, Forest, ASAL regions
Typical hosts	KMD stations, airports, schools, research centres, community institutions
Data platform	CHORDS (Cloud-Hosted Real-time Data Services) Custom Build Dashboard (Under development)

5 DATA INFRASTRUCTURE AND MONITORING SYSTEMS

Data collected from the 3D-PAWS network is transmitted through cellular communication networks to a centralized cloud-based platform, where it is stored and made available for analysis and visualization. The integration of cloud-based infrastructure with customized monitoring systems facilitates efficient management of distributed weather stations and enhances the accessibility and usability of meteorological observations.

5.1 Data Flow Architecture

The 3D-PAWS system employs a hierarchical data transmission architecture in which meteorological observations are acquired by the installed sensors, processed by the station datalogger, and transmitted through cellular communication networks to the CHORDS cloud platform. The observations are subsequently accessed through web-based applications and custom-developed dashboards for real-time monitoring, visualization, and analysis.

The overall data flow can be represented as:



Figure 5:1 : Overall 3D-PAWS data flow architecture.

This architecture enables observations collected at distributed 3D-PAWS stations to be transmitted from the field to a centralized cloud-based platform, where they can be accessed for monitoring and subsequent analysis.

5.2 CHORDS Platform

The Cloud-Hosted Real-time Data Services (CHORDS) platform provides centralized storage, access, and visualization of observations from the 3D-PAWS network. The platform supports both real-time and historical analysis of meteorological observations and enables remote access to data from multiple stations.

The CHORDS platform provides capabilities for:

- Real-time data visualization of observations received from deployed 3D-PAWS stations;
- Historical data retrieval for subsequent analysis and validation;
- Time-series analysis of meteorological parameters to examine temporal variability and observation trends;
- Station management and organization of observations from multiple network locations;
- Monitoring of network connectivity and data transmission, enabling identification of stations experiencing communication or transmission interruptions; and
- Grafana-based visualization and dashboard configuration, providing an additional graphical interface for displaying selected 3D-PAWS observations through configurable panels, time-series plots, and station monitoring views.

The integration of CHORDS with Grafana provides complementary data visualization capabilities. CHORDS serves as the principal platform for receiving, storing, accessing, and managing observations, while the configured Grafana dashboards provide flexible visualization of selected observations and time-series information. This configuration supports both operational monitoring and subsequent analysis of the 3D-PAWS network.

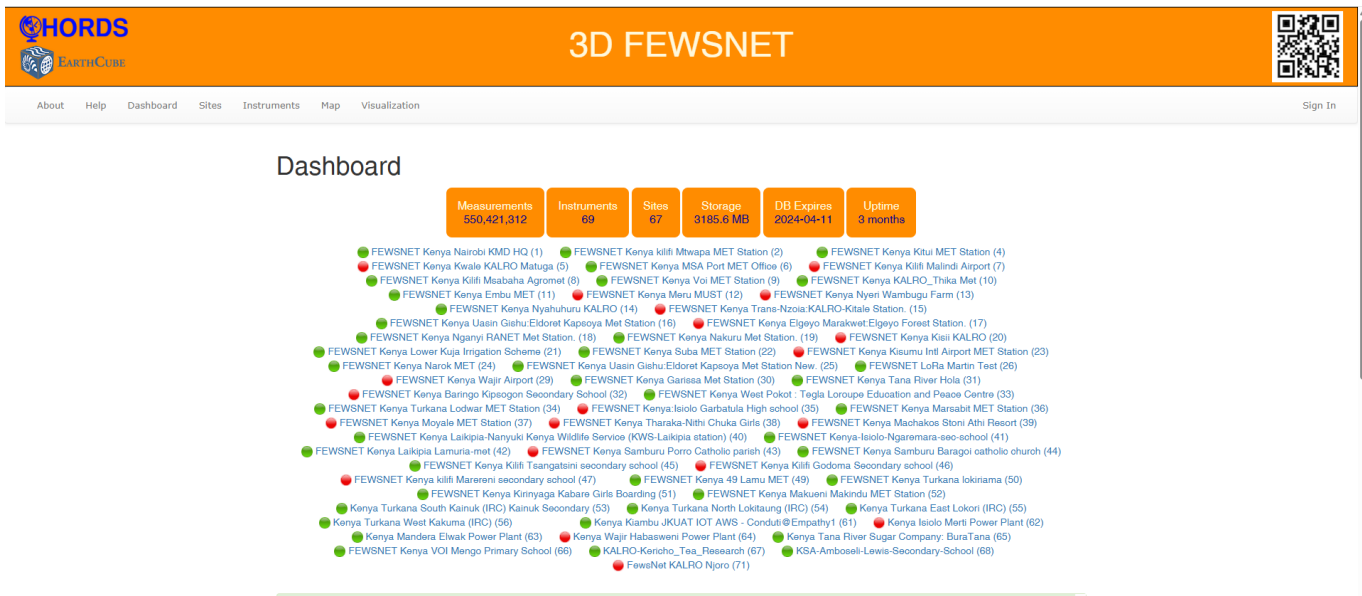


Figure 5:2 Chords Dashboard



Figure 5:3 Grafana Visualization Dashboard

5.3 Custom Dashboard (Under Development)

To improve usability, network monitoring, and access to observations, a custom web-based dashboard is being developed to complement the existing CHORDS data platform, as shown in Figure 5.4. The dashboard is designed to provide a centralized interface for monitoring the 3D-PAWS observation network and accessing meteorological data from multiple stations.

The dashboard will present key network and system indicators, including the number of active and inactive stations, instrument availability, data transmission status, data storage statistics, and overall system uptime. It will also organize stations geographically by operational zones, enabling users to view and manage stations according to their respective zones while retaining station-level access to observations.

A major feature of the dashboard is the ability to access and download summarized station observations over different temporal periods. Users will be able to retrieve station-level summaries at daily, monthly, and other configurable aggregation periods, supporting operational monitoring, climatological analysis, reporting, and further validation.

The major functionalities of the dashboard include:

- Station inventory management and organization of the 3D-PAWS network;
- Zonal organization and visualization of stations to facilitate regional network monitoring;
- Real-time data visualization from active stations;
- Monitoring of active and inactive stations and their data transmission status;
- Instrument availability and operational-status monitoring;
- Data storage and observation statistics;
- Network connectivity and uptime monitoring;
- Station-level meteorological data visualization and analysis;
- Daily, monthly, and configurable-period data summarization;
- Download of summarized station observations for analysis and reporting; and
- User management and controlled access to dashboard functions.

The dashboard therefore complements CHORDS by providing a more user-oriented interface for network management, visualization, analysis, and data access. While CHORDS provides the underlying data platform for receiving and accessing observations, the custom dashboard is intended to provide tailored tools for KMD operational users, including zonal monitoring and simplified retrieval of summarized observations.

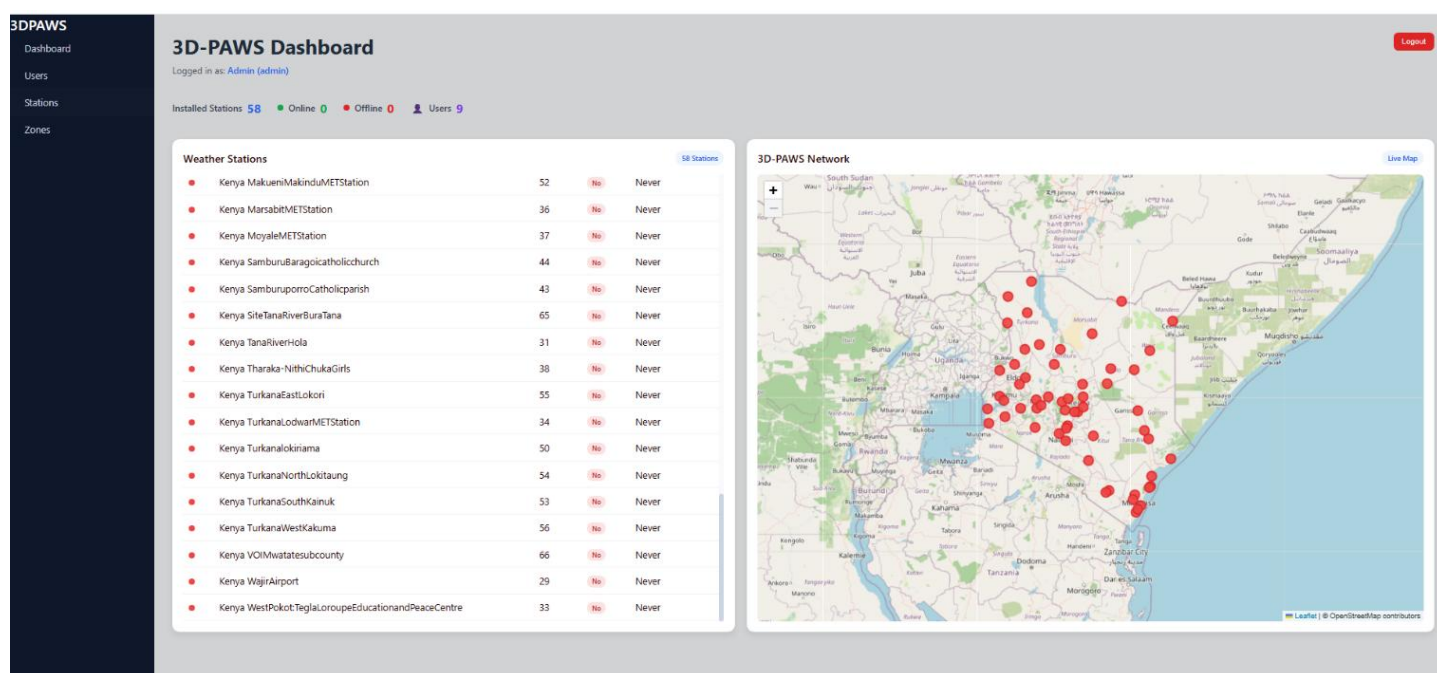


Figure 5:4 custom 3D-PAWS Dashboard

5.3.1 Station Health Monitoring

The custom dashboard incorporates station health monitoring capabilities designed to assess the operational status and connectivity of individual 3D-PAWS stations. The monitoring indicators provide users with an overview of station condition and data transmission performance and support early identification of potential operational problems.

The key indicators monitored include:

- Battery voltage, to assess the power status of the station;

- Cellular signal strength, to indicate the quality of network connectivity;
- Data transmission delays, to identify interruptions or delays in the delivery of observations;
- Communication status, to determine whether a station is successfully transmitting data;
- Sensor availability, to identify sensors that are not providing observations; and
- Instrument uptime, to assess the operational availability of the station over time.

These indicators enable users to identify stations requiring attention without having to inspect each station Physically. The health-monitoring functionality therefore supports early fault detection, network-wide operational awareness, and more efficient maintenance planning.

5.3.2 Geospatial Monitoring

The custom dashboard being developed incorporates geospatial visualization capabilities to provide a spatial representation of the 3D-PAWS observation network. Interactive mapping will enable users to visualize station locations, assess network distribution and coverage, and monitor the operational status of stations across the different zones.

The geospatial interface will allow users to identify individual stations and access relevant station information, including operational status and data availability. The zonal organization of stations will further support monitoring of geographically distributed observation sites and facilitate assessment of network coverage.

These capabilities are intended to support efficient management of the distributed observation network, improve situational awareness, and provide users with a consolidated spatial view of station deployment and operational performance.

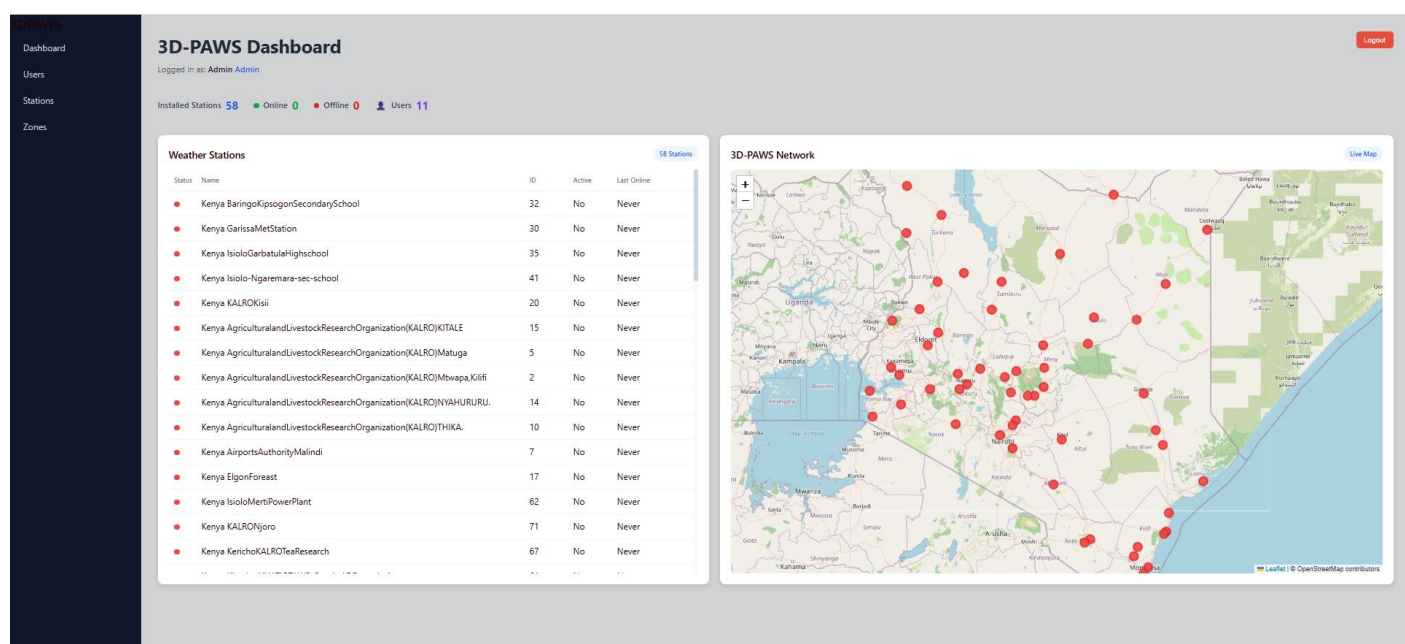


Figure 5:5 Geospatial Distribution Dashboard

5.3.3 Data Availability Monitoring

The custom dashboard will also incorporate data availability monitoring to support assessment of the reliability and operational performance of the 3D-PAWS observation network. Continuous monitoring of incoming observations will enable users to identify periods of missing data, delayed observations, and potential transmission failures.

Data availability indicators and station-level statistics will be used to assess the continuity of observations and identify stations requiring further investigation or maintenance. The dashboard will provide these indicators alongside station health and communication information, allowing data availability problems to be considered together with the operational status of individual stations.

Monitoring data availability will support:

- Identification of missing or interrupted observations;
- detection of data transmission failures and delays and assessment of station-level data continuity;
- comparison of data availability across stations and zones;
- identification of stations requiring maintenance or technical attention; and
- evaluation of the overall reliability and operational performance of the observation network.

This functionality will contribute to more effective maintenance planning and support the long-term sustainability and expansion of the 3D-PAWS observation network.

The dashboard architecture is also designed to support future integration with advanced analytical and decision-support applications. This provides a foundation for extending the system beyond observation visualization and monitoring to support more advanced data analysis, quality-control processes, and operational decision-making as the 3D-PAWS network expands.

The integration of cloud-based data infrastructure, CHORDS, Grafana visualization, and the custom dashboard provides a comprehensive framework for managing, accessing, visualizing, and monitoring observations from the 3D-PAWS network. The approach enhances the accessibility and operational management of distributed observations while providing a foundation for future expansion of the network and integration of additional digital services.

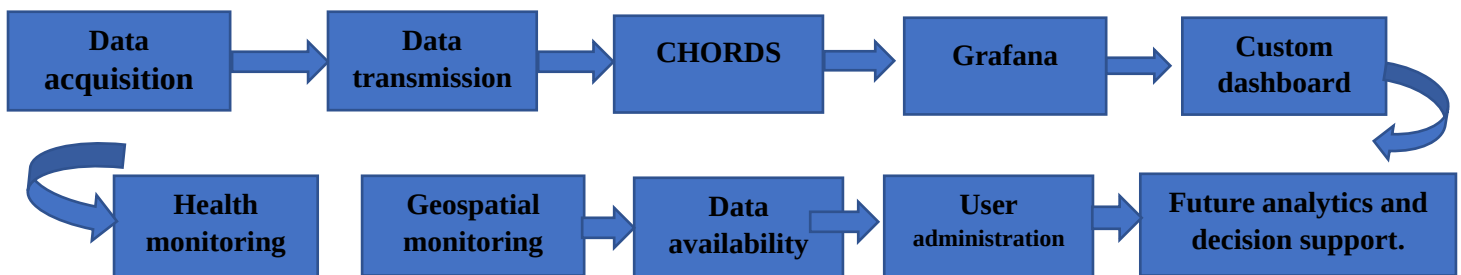


Figure 5:6 3D-PAWS Data Flow

5.3.4 Stations Administration

The custom 3D-PAWS dashboard provides a dedicated Station Administration interface for managing the observation network. The interface provides a centralized station inventory through which authorized users can add, edit, and delete station records and associate individual stations with their corresponding devices and operational zones.

As illustrated in Figure 5.7, the station administration interface captures key metadata for each station, including station name, device identification, geographic coordinates, elevation, assigned zone, and operational status. The interface also provides user assignment functionality, allowing responsibility for individual stations to be associated with designated users or operational teams.

The station inventory further provides an overview of the current status of stations, including identification of stations that are online or offline and a click on the station redirects the user to the stations page. Filtering by operational zone enables users to organize and manage stations according to their geographical or programme-specific classification.

The station administration functionality therefore provides a centralized mechanism for maintaining the 3D-PAWS network inventory and supports:

- Addition and registration of new 3D-PAWS stations;
- Assignment of geographic coordinates, station elevation and operational zones;
- Assignment of responsible users;

- Monitoring of station online/offline status and editing and maintenance of station metadata;
- Removal or decommissioning of station records when required.

This functionality is particularly important as the 3D-PAWS network expands, since it provides a structured mechanism for maintaining an up-to-date inventory of distributed observation stations.

3DPAWS

Dashboard
Users
Stations
Zones

Stations
View and manage all registered 3D-PAWS monitoring stations.

Logout

Manage Stations
Register, assign and manage monitoring stations.

+ Add Station Total Stations **58**

All Zones
Search stations...

Station	Device ID	County	Zone	Assigned To	Status	Last Seen	Actions
Kenya Baringo Kipsogon Secondary School	32	BARINGO	North Rift	Admin	Offline	Never	Edit Delete
Kenya Garissa Met Station	30	GARISSA	North Eastern	Admin	Offline	Never	Edit Delete
Kenya Isiolo Garbatula High School	35	ISIOLO	Eastern	Admin	Offline	Never	Edit Delete
Kenya Isiolo Ngaremaru sec-school	41	ISIOLO	Eastern	Admin	Offline	Never	Edit Delete
Kenya KALRO Kisii	20	KISII	Nyanza	Admin	Offline	Never	Edit Delete
Kenya Agricultural and Livestock Research Organization (KALRO) KITALE	15	BUNGOMA	Western	Admin	Offline	Never	Edit Delete
Kenya Agricultural and Livestock Research Organization (KALRO) Matuga	5	KWALE	Coast	Admin	Offline	Never	Edit Delete

Figure 5:7 Stations Administration Page

5.3.5 User Administration

The custom dashboard incorporates user administration functions to support controlled and secure access to the 3D-PAWS monitoring system. User roles and access privileges will be used to regulate access to station information, network monitoring functions, data summaries, and analytical tools according to user responsibilities.

3DPAWS

Dashboard
Users
Stations
Zones

Users
Logged in as Admin (Admin)

Logout

Manage Users
Create, edit and manage users who can access the 3D-PAWS platform.

+ Add User Total Users: **11** admin@3dpaws.com

Name	Email	Role	Actions
Admin	admin@3dpaws.com	Admin	Edit Delete
arthur	arthurkuria87@gmail.com	Admin	Edit Delete
FewNet	FewNet@gmail.com	Admin	Edit Delete
IRC	IRC@gmail.com	Customer	Edit Delete
Hypertek	Hypertek@gmail.com	Customer	Edit Delete
KALRO Kericho	KALRO@gmail.com	Customer	Edit Delete
admin2	admin2@3dpaws.com	Customer	Edit Delete
New	Newuser@gmail.com	Customer	Edit Delete

Figure 5:8 User Administration Page

5.3.6 Station-Level Monitoring and Visualization

The custom dashboard provides a dedicated station-level interface through which users can access detailed information for individual 3D-PAWS stations. As illustrated in Figure 5.8, the station page consolidates station metadata, operational health information, geographical location, observation access, visualization, reporting, and maintenance functions within a single interface.

The Station Information panel presents key metadata, including the assigned user, geographic coordinates, elevation, county, and operational zone. This provides users with a concise reference for identifying and managing individual observation sites.

The Station Health panel is designed to provide operational information on communication status, battery condition, cellular signal strength, firmware, last communication or observation time, and sensor availability. These indicators support rapid assessment of the operational condition of individual stations and facilitate identification of stations requiring technical attention.

A geospatial location map is also provided on the station page. The map displays the geographical position of the selected station and provides a spatial reference for its location within the wider observation network.

The station interface further incorporates dedicated sections for:

- Live Data, providing access to current meteorological observations;
- Charts, providing graphical visualization of historical observations;
- Reports and Downloads, supporting retrieval of statistical summaries and station data; and
- Maintenance, supporting station-level maintenance and operational management activities.

The station-level interface therefore provides an integrated view of both meteorological observations and station operational status. By combining station metadata, health indicators, geographic information, data visualization, reporting, and maintenance functions, the interface is intended to reduce the need for users to access separate systems when assessing the condition and performance of an individual station.

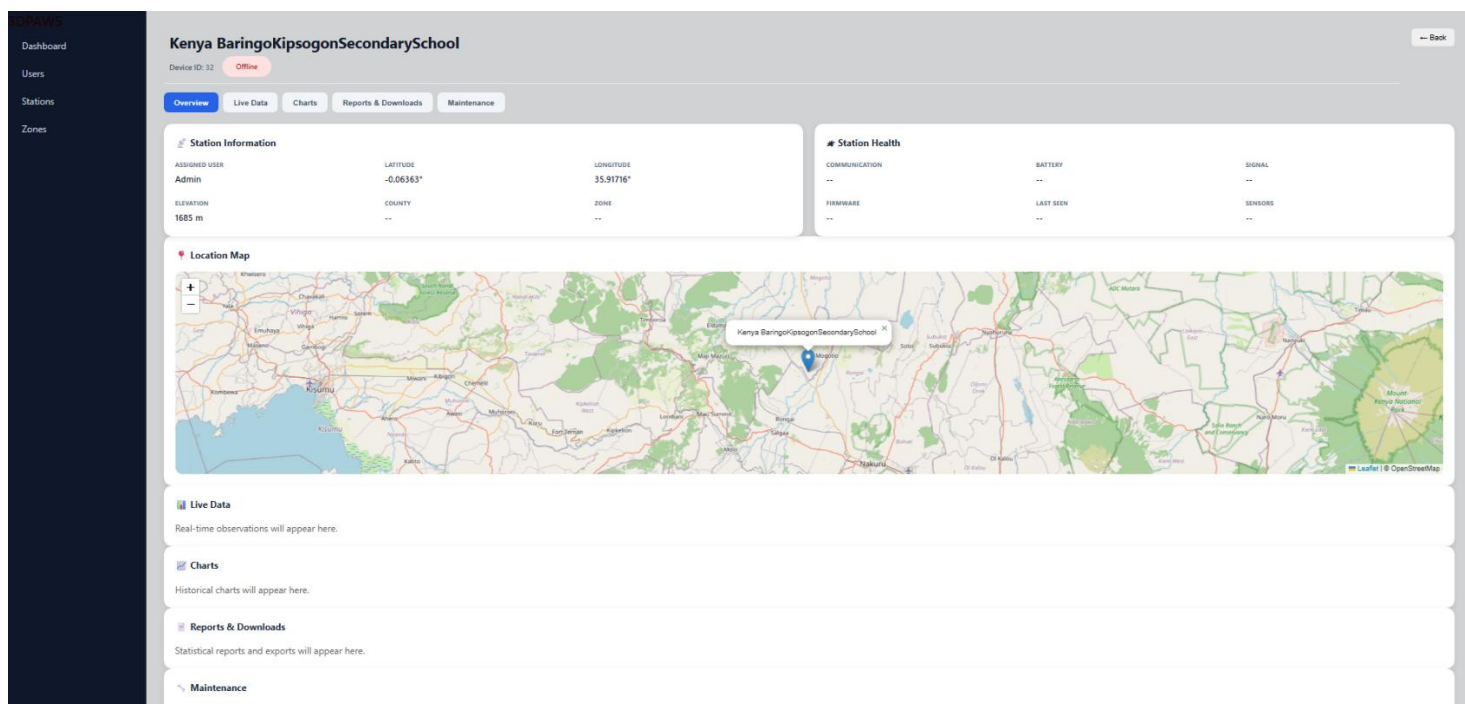


Figure 5:9 Station Level Dashboard view

6.1 Introduction

This chapter will describe the methodology used to evaluate the performance and operational suitability of 3D-Printed Automatic Weather Stations (3D-PAWS) deployed in Kenya. The methodology combines field deployment, observation data acquisition, data quality control, dataset standardization, temporal and spatial comparison with Kenya Meteorological Department (KMD) reference observations, statistical validation, visualization, and performance assessment.

The evaluation is designed to determine the extent to which locally fabricated 3D-PAWS can provide observations that are sufficiently consistent with established meteorological observations for operational and network-densification applications.

Rather than evaluating a single station or a single short observation period, the study adopts a multi-station and multi-environment approach. A target observation period of approximately two to three years was established, subject to the availability and continuity of both 3D-PAWS and corresponding KMD reference observations. The selected stations represent contrasting climatic and agro-ecological environments in Kenya, including high-rainfall, high-altitude, agro-pastoral, marginal-agricultural, pastoral, and coastal environments.

The analysis focuses primarily on temperature, relative humidity, atmospheric pressure and rainfall, with the specific parameters evaluated determined by sensor availability, data quality, and availability of corresponding KMD reference observations.

6.2 Research Design

The study adopts a comparative observational validation design.

The approach involves comparing observations generated by 3D-PAWS with observations from corresponding KMD reference stations. The KMD observations provide the reference dataset against which the performance of the 3D-PAWS observations is evaluated.

The methodology consists of the following major stages:

1. Selection of representative 3D-PAWS stations.
2. Identification of corresponding KMD reference datasets.
3. Acquisition of daily observations covering approximately 2–3 years.
4. Standardization of the 3D-PAWS and KMD datasets.
5. Quality assessment of 3D-PAWS observations.
6. Alignment of the two datasets according to common observation dates.
7. Construction of paired comparison datasets.
8. Calculation of descriptive statistics.
9. Calculation of statistical validation metrics.
10. Assessment against applicable WMO performance criteria.
11. Generation of comparative graphs.
12. Interpretation of results at station, climatic-zone and overall network levels.

This workflow is implemented through the developed 3D-PAWS Validation Toolkit, which provides a reproducible computational framework for processing and analysing the observations

6.3 Study Area and Selection of Observation Sites

The evaluation is designed to cover contrasting climatic and agro-ecological environments within Kenya. Site selection is based on the need to evaluate the performance of 3D-PAWS under different climatic, elevation, rainfall, land-use and operational conditions.

The study targets six representative climatic/agro-ecological zones or regional environments distributed across Kenya. These environments are selected to provide a broad representation of the conditions under which 3D-PAWS has been deployed as part of an expanded meteorological observation network.

Table 6:1 representative climatic/agro-ecological zones or regional environments

Region	Representative Areas	Environmental Characteristic
Western Kenya	Trans Nzoia	High-rainfall environment
Central Kenya	Nyandarua / Nyahururu	High-altitude environment
South Rift	Narok	Agro-pastoral environment
South-Eastern Kenya	Makueni / Kitui	Marginal-agricultural environment
Eastern/Northern Kenya	Marsabit, Garissa, Tana River	Pastoral environment
Coastal Kenya	Kilifi / Taita-Taveta	Coastal environment

The purpose of this distribution is not merely geographic representation. It allows the evaluation to determine whether the performance of 3D-PAWS remains consistent across substantially different environmental conditions.

For each selected location, the corresponding 3D-PAWS station is matched with an appropriate KMD reference station or dataset based on station location, observation period and availability of comparable meteorological parameters.

6.4 Observation Data

6.4.1 3D-PAWS Data

The 3D-PAWS datasets constitute the primary observation data generated by the locally fabricated automatic weather stations.

The stations transmit observations through the deployed data communication infrastructure, with observations subsequently made available through the CHORDS platform and associated data-processing systems.

The raw observations contain multiple sensor parameters and system-status variables. The validation workflow identifies the parameters required for comparison and maps the relevant 3D-PAWS variables to standardized analytical names.

For example, the CHORDS dataset may contain sensor variables such as:

- temperature;
- relative humidity;
- atmospheric pressure;
- rainfall;
- battery and system-status parameters; and
- other derived or diagnostic variables.

The primary comparison variables are selected according to the availability and quality of corresponding observations from the KMD reference dataset.

6.4.2 KMD Reference Data

KMD observations constitute the reference dataset for the comparative evaluation.

The KMD data are standardized into a consistent daily format containing the relevant meteorological parameters and associated station information.

The reference datasets provide the benchmark against which the 3D-PAWS observations are evaluated.

Where possible, the comparison uses:

- the same observation date;
- the same meteorological parameter;
- the corresponding station or representative reference location; and
- the same analysis periods.

This ensures that the statistical comparison is based on paired observations rather than independent datasets.

6.4.3 Observation Period

The study targets daily observations covering approximately two to three years for each selected station, subject to the availability and continuity of corresponding 3D-PAWS and KMD observations. The available datasets for each station will first be assessed to determine the period over which both observing systems provide sufficiently comparable records.

The use of a multi-year period is intended to capture:

- seasonal variability;
- wet and dry seasons;
- inter-seasonal changes;
- periods of high and low temperature;
- periods of high and low humidity;
- rainfall variability;
- operational data gaps; and
- changes in sensor performance over time.

The actual analysis period for each station will depend on the availability and continuity of both 3D-PAWS and KMD observations.

Where the two datasets do not completely overlap, the analysis will use their common observation period rather than artificially extending or filling the datasets.

This principle is important because it prevents the statistical analysis from comparing observations from different periods.

6.5 Data Processing and Standardization

Before statistical analysis, the 3D-PAWS and KMD datasets are processed and standardized into a common analytical structure using a custom-developed 3D-PAWS Validation Toolkit. The toolkit has been developed specifically to support the systematic processing, comparison, quality assessment, statistical analysis, visualization, and reporting of 3D-PAWS observations against corresponding KMD reference observations.

The processing workflow consists of the following stages:

6.5.1 Data Loading

The toolkit automatically identifies and loads the selected 3D-PAWS and KMD datasets for the selected observation station.

The Dataset Manager identifies available stations and associated datasets, while the Comparison Manager identifies compatible 3D-PAWS and KMD datasets based on station, observation period, and dataset availability.

The selected 3D-PAWS dataset is loaded together with the corresponding KMD reference dataset for subsequent processing.

6.5.2 Date Standardization

Observation dates and timestamps from the two data sources are standardized into a common date format.

This step ensures that observations from 3D-PAWS and KMD can be correctly aligned according to their observation dates and prevents differences in date formatting from affecting the comparison.

6.5.3 Dataset Period Alignment

The toolkit determines the period over which the selected 3D-PAWS and KMD datasets have overlapping observations.

The datasets are subsequently restricted to their common observation period to ensure that the statistical comparison is performed using observations representing the same period.

Where the datasets have different start or end dates, observations outside the common period are excluded from the paired comparison rather than being artificially supplemented.

6.5.4 Variable Mapping

The different variable naming conventions used by CHORDS/3D-PAWS and KMD are mapped to a common set of analytical variables.

This mapping allows equivalent meteorological parameters from the two observing systems to be directly compared despite differences in their original variable names and data structures.

The principal variables considered in the comparison include temperature, relative humidity, atmospheric pressure and rainfall.

For example, the toolkit maps the relevant 3D-PAWS and KMD variables into standardized analytical pairs such as:

Table 6:2 standardized analytical pairs

Parameter	3D-PAWS Variable	KMD Variable
Maximum Temperature	Tmax_3D-PAWS	Tmax_KMD
Minimum Temperature	Tmin_3D-PAWS	Tmin_KMD
Maximum Relative Humidity	RH_Max_3D-PAWS	RH_Max_KMD
Minimum Relative Humidity	RH_Min_3D-PAWS	RH_Min_KMD
Mean Relative Humidity	RH_Mean_3D-PAWS	RH_Mean_KMD
Maximum Pressure	Pressure_Max_3D-PAWS	Pressure_Max_KMD
Minimum Pressure	Pressure_Min_3D-PAWS	Pressure_Min_KMD
Mean Pressure	Pressure_Mean_3D-PAWS	Pressure_Mean_KMD
Rainfall — Gauge 1	Rain_RG1_3D-PAWS	Rain_RG1_KMD
Rainfall — Gauge 2	Rain_RG2_3D-PAWS	Rain_RG2_KMD

The specific variables included in the final analysis depend on sensor availability, data quality, and the availability of corresponding KMD observations for each station.

This standardized structure provides the basis for the subsequent data-quality assessment, paired observation analysis, statistical validation, graphical comparison, and performance assessment described in the following sections.

6.6 Data Quality Control

Quality control was applied to the meteorological observations before statistical comparison between the 3D-PAWS and KMD reference datasets. The quality-control procedures were parameter-specific to account for the physical characteristics of each meteorological variable and to avoid incorrectly removing valid observations.

The developed toolkit therefore distinguishes between:

- Valid observations
- Invalid/faulty observations
- Missing observations

Variable-specific quality-control rules are applied.

6.6.1 Temperature

Temperature observations were screened for invalid values according to the quality-control rules implemented in the validation toolkit. Observations failing the defined validity criteria were excluded from subsequent paired analysis, while valid observations were retained for comparison with the KMD reference dataset.

6.6.2 Relative Humidity

Relative humidity observations were evaluated against the physically meaningful range of 0–100%. Observations falling outside this range were classified as invalid and excluded from the subsequent statistical analysis.

6.6.3 Atmospheric Pressure

Atmospheric pressure observations were screened against the defined operational range implemented in the validation toolkit. Values outside the specified range were treated as invalid and excluded from the paired-observation analysis.

6.6.4 Rainfall

Rainfall observations required separate treatment because zero rainfall is a valid meteorological observation rather than an indication of missing or faulty data.

A rainfall value of 0 mm indicates that no rainfall was recorded during the corresponding observation period and was therefore retained as a valid observation. Under the quality-control rules implemented in the validation toolkit, negative rainfall values were classified as invalid.

This distinction is particularly important for rainfall validation because zero-rainfall observations contribute directly to the representation of rainfall occurrence and dry periods. Removing valid zero observations could therefore introduce bias into rainfall occurrence, frequency, accumulation, and agreement statistics. Consequently, zero rainfall values were retained in the analysis, while only observations failing the defined rainfall validity rule were excluded.

6.7 Sensor Quality and Data Availability Assessment

The methodology distinguishes between sensor quality and comparison-data coverage.

A station may have a functioning sensor but still have incomplete comparison data because one of the datasets contains missing dates.

Therefore, the toolkit reports:

- total observations;
- valid observations;
- faulty observations;
- missing observations;
- paired observations;
- comparison-period missing observations; and
- overall data-quality status.

The current classification includes conditions such as:

- READY
- PARTIAL COVERAGE
- QUALITY ISSUES
- FAULTY SENSOR
- NO PAIRED DATA
- MISSING

A sensor is not classified as a failed sensor merely because some observations are missing or faulty.

The current toolkit uses the proportion of explicitly classified faulty observations to identify severe sensor failure. A sensor is classified as FAULTY SENSOR when at least 90% of its classified observations are explicitly faulty.

This threshold is applied consistently across the validation workflow.

6.8 Construction of Paired Comparison Dataset

After quality control and temporal alignment, the toolkit constructs a paired comparison dataset.

For each parameter, a pair consists of:

one valid 3D-PAWS observation and the corresponding KMD observation for the same date.

Observations for which either dataset lacks a usable value are excluded from the statistical pairing for that parameter.

Thus: $N_{\text{paired}} = N(\text{valid 3D-PAWS and valid KMD observations})$

This ensures that the calculated validation statistics represent direct comparisons between the two observing systems.

The comparison dataset also preserves dates where one or both observations are unavailable so that data coverage and missingness can be reported separately from statistical performance.

6.9 Descriptive Statistical Analysis

Descriptive statistics are first calculated independently for 3D-PAWS and KMD.

The following statistics are calculated:

- number of observations;
- mean & median;
- minimum & maximum;
- standard deviation; and variance.

These statistics provide an initial assessment of whether the distributions and ranges of 3D-PAWS observations are comparable to those of the KMD reference observations.

For rainfall, daily totals are used to characterize rainfall occurrence and accumulation

6.10 Statistical Validation

The performance of 3D-PAWS is evaluated using paired statistical measures.

6.10.1 Bias

Bias measures the mean difference between the 3D-PAWS and KMD observations:

$\text{Bias} = \frac{1}{n} \sum_{i=1}^n (P_i - K_i)$:where:

- P_i is the 3D-PAWS observation;
- K_i is the KMD observation; and
- n is the number of paired observations.

A positive bias indicates that 3D-PAWS tends to record higher values than KMD, while a negative bias indicates lower values.

6.10.2 Mean Absolute Error

Mean Absolute Error (MAE) is calculated as: $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |P_i - K_i|$

MAE provides the average magnitude of the difference between the two observing systems without considering the direction of the error.

6.10.3 Root Mean Square Error

Root Mean Square Error (RMSE) is calculated as: $\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (P_i - K_i)^2}$

RMSE gives greater weight to larger errors and is therefore useful for identifying periods where the 3D-PAWS observation differs substantially from the KMD reference.

6.10.4 Correlation

The Pearson correlation coefficient is calculated to evaluate the degree to which the temporal variability in 3D-PAWS observations follows the corresponding KMD observations.

$$r = \frac{\sum (P_i - \bar{P})(K_i - \bar{K})}{\sqrt{\sum (P_i - \bar{P})^2 \sum (K_i - \bar{K})^2}}$$

A high positive correlation indicates that the 3D-PAWS observations reproduce the temporal variability observed by the KMD reference station.

Correlation is interpreted together with error statistics because a high correlation does not necessarily imply low measurement error.

6.10.5 Coefficient of Determination

The coefficient of determination is calculated as:

$$R^2=r^2$$

R^2 provides an indication of the proportion of variance represented by the linear relationship between the 3D-PAWS and KMD observations.

6.11 Graphical Analysis

Statistical results are complemented by graphical analysis.

The developed plotting module automatically generates station-specific graphs for each usable parameter.

6.11.1 Time-Series Comparison

Time-series graphs show 3D-PAWS and KMD observations against observation date.

These graphs are used to examine:

- seasonal patterns;
- temporal agreement;
- systematic differences;
- extreme observations;
- periods of divergence; and
- data gaps.

6.11.2 Scatter Comparison

Scatter plots compare 3D-PAWS observations directly against KMD observations.

These plots provide a visual assessment of:

- linear agreement;
- dispersion;
- systematic bias;
- outliers; and
- consistency across the observation range.

6.11.3 Rainfall Accumulation

For rainfall, cumulative rainfall plots are additionally generated.

These allow comparison of accumulated rainfall recorded by 3D-PAWS and KMD over the observation period.

6.12 WMO Performance Assessment

Where an applicable WMO performance requirement is available for the parameter, the calculated validation statistics are assessed against the relevant WMO criteria.

The toolkit distinguishes between:

- Goal
- Breakthrough
- Threshold
- Below Threshold

The performance classification is based on the applicable error criterion and is considered together with the strength of the correlation.

Parameters that do not contain sufficient valid paired observations or are classified as having a failed sensor are not formally assessed.

This prevents inadequate data from being interpreted as evidence of poor or good sensor performance.

6.13 Multi-Station and Environmental Comparison

A major component of the study is comparison of 3D-PAWS performance across the six representative environmental regions.

The station-level analysis will subsequently be aggregated by environmental zone:

Table 6:3 Representative stations in Regions

ZONE	REGION	Representative station
Zone 1	Western High-Rainfall Environment	Trans Nzoia, Kitale KARLO Met
Zone 2	Central High-Altitude Environment	Nyandarua / Nyahururu Met
Zone 3	South Rift Agro-Pastoral Environment	Narok MET Station
Zone 4	South-Eastern Marginal-Agricultural Environment	Kitui Met Station
Zone 5	Eastern/Northern Pastoral Environment	Garissa Met Station
Zone 6	Coastal Environment	Kilifi Msabaha / Mtwapa Met

The purpose of this analysis is to determine whether 3D-PAWS performance is consistent across different environmental conditions.

The comparison will consider:

- bias;
- MAE;
- RMSE;
- correlation;
- R^2 ;
- data availability;
- faulty observations;
- missing observations;
- WMO performance classification; and
- rainfall accumulation agreement where applicable.

This will allow the study to determine whether observed performance is generally consistent or whether certain environmental conditions present particular challenges to the 3D-PAWS design.

6.14 Data Completeness and Multi-Year Assessment

Because the target dataset covers approximately 2–3 years, data completeness will also be evaluated.

For each station, the analysis will document:

- $\text{Data Completeness} = \frac{\text{Expected Observations}}{\text{Valid Observations}} \times 100$

The completeness assessment will be interpreted alongside the statistical performance.

This distinction is important because:

A sensor may have excellent agreement when observations are available but still have poor operational availability if substantial periods of data are missing.

Consequently, the study will assess both:

1. measurement performance, and
2. data availability/reliability.

6.15 Validation Toolkit Architecture

The developed 3D-PAWS Validation Toolkit operationalizes the validation methodology described in this study as a structured and reproducible analytical workflow. The toolkit was designed to support the processing, comparison, statistical evaluation and performance assessment of observations obtained from 3D-PAWS stations against corresponding reference observations. Its modular architecture allows individual processing functions to be maintained independently while enabling the complete validation workflow to be executed consistently across multiple stations and observation periods.

The workflow begins with dataset discovery and metadata inspection, followed by dataset selection, temporal alignment and preparation of paired observations. The 3D-PAWS observations are processed through the CHORDS data-processing component, after which quality-control procedures and statistical calculations are applied. The resulting statistics are subsequently evaluated against the defined WMO performance criteria. Finally, the toolkit generates graphical and tabular outputs that support interpretation of station-level and network-level performance.

The principal processing modules and their methodological functions are summarized in **Table 6.4**.

The main processing components are:

Table 6:4 Validation Toolkit Architecture

Module	Methodological Function
dataset_manager.py	Identifies available stations and datasets
dataset_info.py	Reads dataset metadata and quality information
comparison_manager.py	Selects, aligns and merges datasets
comparison_job.py	Stores the selected comparison workflow
chords_processor.py	Processes 3D-PAWS/CHORDS observations
statistics_utils.py	Performs observation-quality filtering and statistical calculations
statistical_analysis.py	Performs descriptive and validation analysis
validation_statistics.py	Supports validation metrics
performance_assessment.py	Applies WMO performance criteria
plotting.py	Generates validation graphs & reports

The modular structure provides a clear separation between data preparation, quality control, statistical analysis, performance assessment and visualization. This separation reduces the need for manual intervention and improves consistency when processing datasets from different stations. It also enables the same analytical procedure to be applied to different stations and observation periods without modifying the underlying statistical methodology.

The architecture further supports the transition from station-level validation to network-level assessment. Individual station datasets can first be processed using the same workflow, after which the resulting statistics can be consolidated to produce network-wide summaries. This approach was subsequently used in Chapter 7 to combine the results from the eight evaluated stations and generate the consolidated validation dataset comprising 79 parameter-level assessments.

The toolkit therefore serves not only as a computational tool but also as a mechanism for ensuring reproducibility, consistency and traceability of the validation process. The generated statistical tables and graphical outputs provide the basis for evaluating the performance of the 3D-PAWS network against the selected KMD reference observations and the applicable WMO performance criteria.

6.16 Results Reporting

The final outputs of the methodology are generated for each selected station and comparison period.

The toolkit produces:

1. standardized comparison datasets;
2. data-quality summaries;
3. descriptive statistics;
4. validation statistics;
5. WMO performance assessments;
6. time-series plots;
7. scatter plots;
8. rainfall accumulation plots;
9. station-level interpretations; and

The generated report is intended to provide a structured presentation of the results in the following chapter

6.17 Quality Assurance and Reproducibility

To improve reproducibility, the same processing and analytical procedures are applied to all selected stations.

The toolkit records the selected station, datasets, comparison period, paired observations, data-quality status, statistical results and performance assessment.

The methodology therefore minimizes manual manipulation of the datasets and ensures that the same statistical procedures are applied consistently across stations.

This is particularly important for the multi-station evaluation because differences between stations should primarily reflect differences in observations and environmental conditions rather than differences in analytical procedures.

6.18 Summary

The methodology establishes a multi-station comparative framework for evaluating 3D-PAWS observations against KMD reference observations across contrasting Kenyan climatic and agro-ecological environments. Daily observations covering approximately two to three years are targeted to capture seasonal and inter-annual variability.

The developed validation toolkit provides an integrated workflow for data ingestion, quality control, temporal alignment, paired dataset construction, statistical analysis, graphical comparison, WMO performance assessment and automated results reporting. Particular attention is given to distinguishing valid zero rainfall observations from missing or faulty observations.

7 RESULTS AND VALIDATION ASSESSMENT

This consolidated chapter presents the results of the comparative validation of eight 3D-PAWS stations against corresponding Kenya Meteorological Department (KMD) reference observations. The chapter retains the station-level statistical evidence generated by the 3D-PAWS Validation Toolkit and integrates it into a network-level assessment. Particular attention is given to observation quality, paired-data coverage, bias, mean absolute error (MAE), root mean square error (RMSE), correlation coefficient (r), coefficient of determination (R^2), and the performance classifications implemented in the toolkit.

The consolidated presentation does not treat the performance classification as a standalone measure of operational suitability. Correlation is interpreted as evidence of temporal agreement, while bias, MAE and RMSE describe numerical agreement. Observation coverage and data quality are explicitly considered because several stations contain substantial differences in the number of paired days available for different parameters.

Table 7:1 Summary of statistical indicators that should be used together with the achieved results

No	Indicator	Interpretation
1	Bias	A positive bias indicates that 3D-PAWS tends to record higher values than KMD, while a negative bias indicates lower values.
2	Mean Absolute Error (MAE)	Provides the average magnitude of the difference between the two observing systems without considering the direction of the error.
3	Root Mean Square Error (RMSE)	Gives greater weight to larger errors and is therefore useful for identifying periods where the 3D-PAWS observation differs substantially from the KMD reference.
4	Correlation (r)	A high positive correlation indicates that the 3D-PAWS observations reproduce the temporal variability observed by the KMD reference station. Correlation is interpreted together with error statistics because a high correlation does not necessarily imply low measurement error.
5	Coefficient of Determination ($R^2=r^2$)	R^2 provides an indication of the proportion of variance represented by the linear relationship between the 3D-PAWS and KMD observations.

7.1 Overview of the Results

The validation dataset comprised eight co-located 3D-PAWS and KMD station pairs:

- Eldoret
- Garissa
- Kitale-New
- Kitui
- Msabaha Agromet Station
- Mtwapa Agromet Station
- Narok
- Nyahururu Agromet Station.

The available validation periods range from August 2024 to July 2026, with Garissa ending in May 2026 and Kitale-New beginning in October 2024. The station set provides representation across the broad geographical and agro-environmental zones identified for the study:

- Western
- Trans-Nzoia high-rainfall
- Central Nyandarua/Nyahururu high-altitude

- South Rift Narok agro-pastoral
- South East Makueni/Kitui marginal-agricultural
- Eastern Marsabit/Garissa/Tana River pastoral
- Coastal Kilifi/Taita-Taveta coastal.

Eldoret is retained as a separate Rift Valley representation because it is not explicitly named within the six zone labels shown in the attached zoning scheme.

Across the eight station reports, 79 parameter-level assessments are formally generated. Of these, 5 are classified as Breakthrough, 13 as Threshold, and 61 as Below Threshold under the criteria implemented in the validation toolkit. These counts are descriptive of the formal statistical assessments and must be interpreted alongside data quality, paired-observation coverage and the intended operational use of each parameter.

The represented-zone column identifies the broad geographical and environmental setting represented by each validation site. These groupings are used to demonstrate network diversity and are not intended as formal climatic classifications.

Table 7:2 Validation scope and parameter coverage by station

3D-PAWS Station	Kmd Reference Station	Represented Zone	Validation period	Days	Parameters assessed
Eldoret	Eldoret Meteorological Station	Rift Valley / Eldoret	2024-08_to_2026-07	718	11
Garissa	Garissa Meteorological Station	Eastern – Marsabit / Garissa / Tana River Pastoral	2024-08_to_2026-05	642	11
Kitale - New	Kitale Meteorological Station - New	Western – Trans-Nzoia High Rainfall	2024-10_to_2026-07	658	11
Kitui	Kitui Meteorological Station	South East – Makueni / Kitui Marginal-Agricultural	2024-08_to_2026-07	718	5
Msabaha Agromet Station	Msabaha Agromet Station	Coastal – Kilifi / Taita-Taveta Coastal	2024-08_to_2026-07	718	11
Mtwapa Agromet Station	Mtwapa Agromet Station	Coastal – Kilifi / Taita-Taveta Coastal	2024-08_to_2026-07	718	11
Narok	Narok Meteorological Station	South Rift – Narok Agro-Pastoral	2024-08_to_2026-07	718	11
Nyahururu Agromet Station	Nyahururu Agromet Station	Central – Nyandarua / Nyahururu High Altitude	2024-08_to_2026-07	718	8

7.2 Dataset Availability, Completeness and Quality Assessment

Sensor-level quality control was applied using the variable-specific rules implemented in the validation toolkit. Temperature, relative humidity and pressure are reported through the sensor-level quality assessment because continuous sensor observations and QC status are available for these variables. Rainfall is treated separately because the validation dataset contains two tipping-bucket rainfall channels (Rain Gauge 1 and Rain Gauge 2), for which zero values represent valid dry-day observations rather than missing data. Consequently, rainfall quality is evaluated using paired-day coverage, 3D-PAWS rain-gauge consistency and comparison with the KMD co-located gauge.

The network shows substantial differences in data quality and data availability. Temperature, humidity and pressure are reported using sensor-level validity, while rainfall is reported using paired-day coverage because

valid zero rainfall observations represent genuine dry conditions rather than missing data. The combined presentation therefore distinguishes continuous sensor validity from rainfall observation coverage. Garissa, Eldoret and Nyahururu generally show high sensor-level validity for temperature and humidity, whereas Kitui has lower validity across the three continuous sensor groups. Msabaha shows high temperature validity but lower humidity and pressure validity, while Mtwapa shows acceptable temperature and pressure validity with high humidity validity. The rainfall coverage ranges from 25.2% at Kitale-New to 99.1% at Garissa and is interpreted separately from sensor-level validity.

Table 7:3 Sensor-level validity and rainfall data coverage (%) across the eight stations

Station	Temperature	Humidity	Pressure	Rainfall coverage
ELDORET	99.96%	99.96%	99.96%	89.4%
GARISSA	100.00%	100.00%	98.97%	99.1%
KITALE - NEW	99.98%	99.98%	99.94%	25.2%
KITUI	87.66%	87.30%	87.65%	92.8%
MSABAHA AGROMET STATION	99.75%	86.99%	89.94%	94.7%
MTWAPA AGROMET STATION	94.29%	100.00%	94.29%	86.1%
NAROK	92.39%	100.00%	100.00%	94.6%
NYAHURURU AGROMET STATION	100.00%	100.00%	91.43%	82.5%

Table 7:4 Range of daily paired-observation coverage

Station	Lowest coverage	Highest coverage	Parameters
ELDORET	74.9%	89.4%	11
GARISSA	88.0%	99.1%	11
KITALE - NEW	21.7%	25.2%	11
KITUI	27.0%	92.8%	5
MSABAHA AGROMET STATION	39.6%	94.7%	11
MTWAPA AGROMET STATION	0.7%	86.1%	11
NAROK	82.0%	94.6%	11
NYAHURURU AGROMET STATION	37.6%	82.5%	8

The coverage results show that Garissa had the strongest overall daily coverage, ranging from 88.0% to 99.1%, while Kitale-New had very low paired-day coverage, ranging from 21.7% to 25.2%. Mtwapa also contained a major coverage limitation, with pressure coverage extending as low as 0.7%. Kitui had relatively strong rainfall coverage but much lower temperature coverage. These differences are important when comparing stations because a correlation calculated from a small number of paired days is less informative than a similar correlation supported by a long and continuous record.

7.3 Rainfall Data Availability and Internal Gauge Consistency

Rainfall was evaluated separately from the continuous sensor-quality assessment because the 3D-PAWS rainfall system consists of two tipping-bucket gauges. Zero rainfall values are retained as valid observations because they represent dry conditions rather than missing data. The principal data-quality indicators for rainfall are therefore paired-day coverage, the number of available observations and consistency between Rain Gauge 1 and Rain Gauge 2. This approach also allows potential mechanical or recording problems in one gauge to be distinguished from broader station data-transmission problems.

Table 7:5 Rainfall data availability and paired-observation coverage

Station	Rainfall coverage	Paired days	Missing days	Parameters	Data status
ELDORET	89.4%	642	76	11	AVAILABLE
GARISSA	99.1%	636	6	11	AVAILABLE
KITALE - NEW	25.2%	166	492	11	AVAILABLE
KITUI	92.8%	666	52	5	AVAILABLE
MSABAHA AGROMET STATION	94.7%	680	38	11	AVAILABLE
MTWAPA AGROMET STATION	86.1%	618	100	11	AVAILABLE
NAROK	94.6%	679	39	11	AVAILABLE
NYAHURURU AGROMET STATION	82.5%	592	126	8	AVAILABLE

Rainfall paired-day coverage was highest at Garissa (99.1%) and lowest at Kitale-New (25.2%). Eldoret, Kitui, Msabaha, Mtwapa, Narok and Nyahururu provided substantially more rainfall comparison days. The rainfall records are classified as AVAILABLE rather than using the temperature/humidity/pressure sensor-quality categories. This distinction is important because rainfall quality is strongly influenced by gauge mechanics, event characteristics and data availability.

The two-gauge configuration also provides an internal rainfall quality-control check. The supporting rainfall analysis reported Rain Gauge 1 versus Rain Gauge 2 correlations of 0.68 for Kitui, 1.00 for Msabaha, 0.99 for Mtwapa, 0.81 for Narok and 1.00 for Nyahururu. These results indicate very strong internal consistency at Msabaha, Mtwapa and Nyahururu, moderate consistency at Narok, and lower—but still informative—agreement at Kitui. The internal gauge comparison is therefore useful for identifying stations where one rainfall gauge may require inspection before interpreting its comparison with KMD.

7.4 Validation Statistics and Performance Assessment

The validation assessment uses five principal statistical indicators. Bias represents the mean systematic difference between 3D-PAWS and KMD observations. MAE represents the average magnitude of absolute differences, while RMSE gives greater weight to larger discrepancies. The correlation coefficient (r) measures the strength of the linear temporal relationship, and R² expresses the proportion of variance represented by the corresponding linear relationship.

The performance classifications in the station reports are generated using parameter-specific criteria implemented in the validation toolkit. For temperature, the reported Goal, Breakthrough and Threshold values are 0.500 °C, 1.000 °C and 2.000 °C respectively. For relative humidity they are 0.500%, 2.500% and 5.000%; for pressure 0.500 hPa, 1.000 hPa and 1.000 hPa; and for rainfall 0.500 mm, 2.000 mm and 5.000 mm. These values are reported here as the criteria used by the toolkit rather than as an independent reinterpretation of WMO requirements.

Table 7:6 Cross-station performance and strongest temporal agreement by parameter group

Station	Breakthrough	Threshold	Below threshold	Temperature	Humidity	Pressure	Rainfall
Eldoret	0	3	8	Maximum Temperature: r=0.723	Mean Relative Humidity: r=0.912	Maximum Pressure: r=0.332	Daily Rainfall (Rain Gauge 1): r=0.790
Garissa	2	1	8	Maximum Temperature: r=0.940	Mean Relative Humidity: r=0.888	Mean Pressure: r=0.678	Daily Rainfall (Rain Gauge 1): r=0.536
Kitale - New	0	0	11	Maximum Temperature: r=0.616	Mean Relative Humidity: r=0.833	Minimum Pressure: r=0.570	Daily Rainfall (Rain Gauge 2): r=0.142
Kitui	0	0	5	Mean Temperature: r=0.533	Not assessed	Not assessed	Daily Rainfall (Rain Gauge 1): r=0.636
Msabaha Agromet Station	2	1	8	Maximum Temperature: r=0.889	Mean Relative Humidity: r=0.814	Maximum Pressure: r=0.285	Daily Rainfall (Rain Gauge 1): r=0.500
Mtwapa Agromet Station	1	2	8	Mean Temperature: r=0.869	Mean Relative Humidity: r=0.714	Minimum Pressure: r=0.924	Daily Rainfall (Rain Gauge 1): r=0.506
Narok	0	3	8	Minimum Temperature: r=0.892	Mean Relative Humidity: r=0.891	Maximum Pressure: r=0.794	Daily Rainfall (Rain Gauge 1): r=0.618
Nyahururu Agromet Station	0	3	5	Minimum Temperature: r=0.771	Mean Relative Humidity: r=0.923	Not assessed	Daily Rainfall (Rain Gauge 1): r=0.697

7.5 Temperature Results

Temperature was generally the strongest-performing parameter group across the network. Several stations achieved Threshold or Breakthrough classifications, with particularly strong agreement at Garissa, Msabaha, Mtwapa, Narok and Nyahururu. The contrasting Kitui and Kitale-New results demonstrate the importance of data quality and coverage when assessing station performance.

Table 7:7 Cross-station temperature performance highlights

Station	Highest-r parameter	Highest r	Lowest-RMSE parameter	Lowest RMSE	Assessment	Data Status
Eldoret	Maximum Temperature	0.723	Maximum Temperature	1.306	Threshold	High Quality
Garissa	Maximum Temperature	0.940	Maximum Temperature	0.776	Breakthrough	High Quality
Kitale - New	Maximum Temperature	0.616	Mean Temperature	2.085	Below Threshold	High Quality
Kitui	Mean Temperature	0.533	Mean Temperature	2.129	Below Threshold	Limited Quality

Msabaha Agromet Station	Maximum Temperature	0.889	Mean Temperature	0.889	Breakthrough	High Quality
Mtwapa Agromet Station	Mean Temperature	0.869	Mean Temperature	0.760	Breakthrough	Acceptable Quality
Narok	Minimum Temperature	0.892	Minimum Temperature	1.277	Threshold	Acceptable Quality
Nyahururu Agromet Station	Minimum Temperature	0.771	Mean Temperature	1.465	Threshold	High Quality

Garissa is shown because it provides a strong, well-covered temperature comparison, while Kitui illustrates a lower-quality and lower-correlation case.

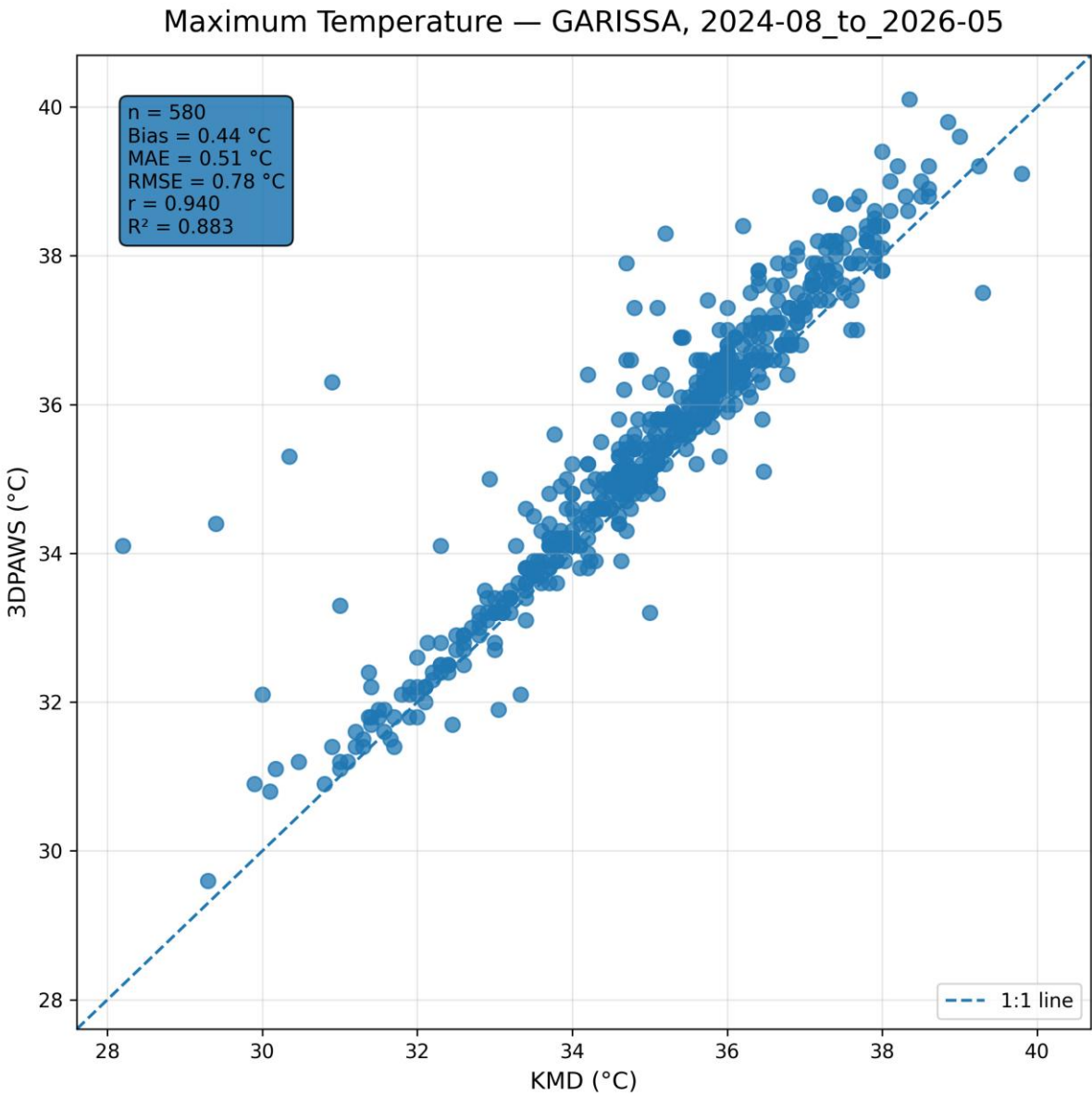


Figure 7:1 scatter comparison for Garissa — representative temperature validation result.

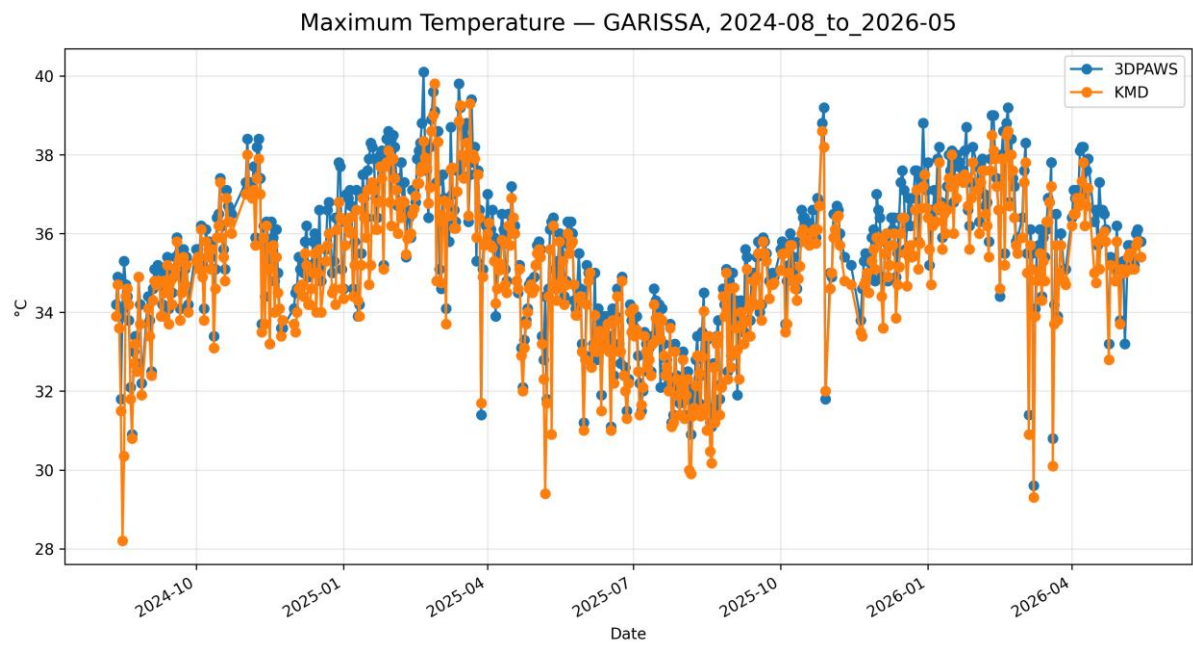


Figure 7:2 time-series comparison for Garissa — representative temperature validation result

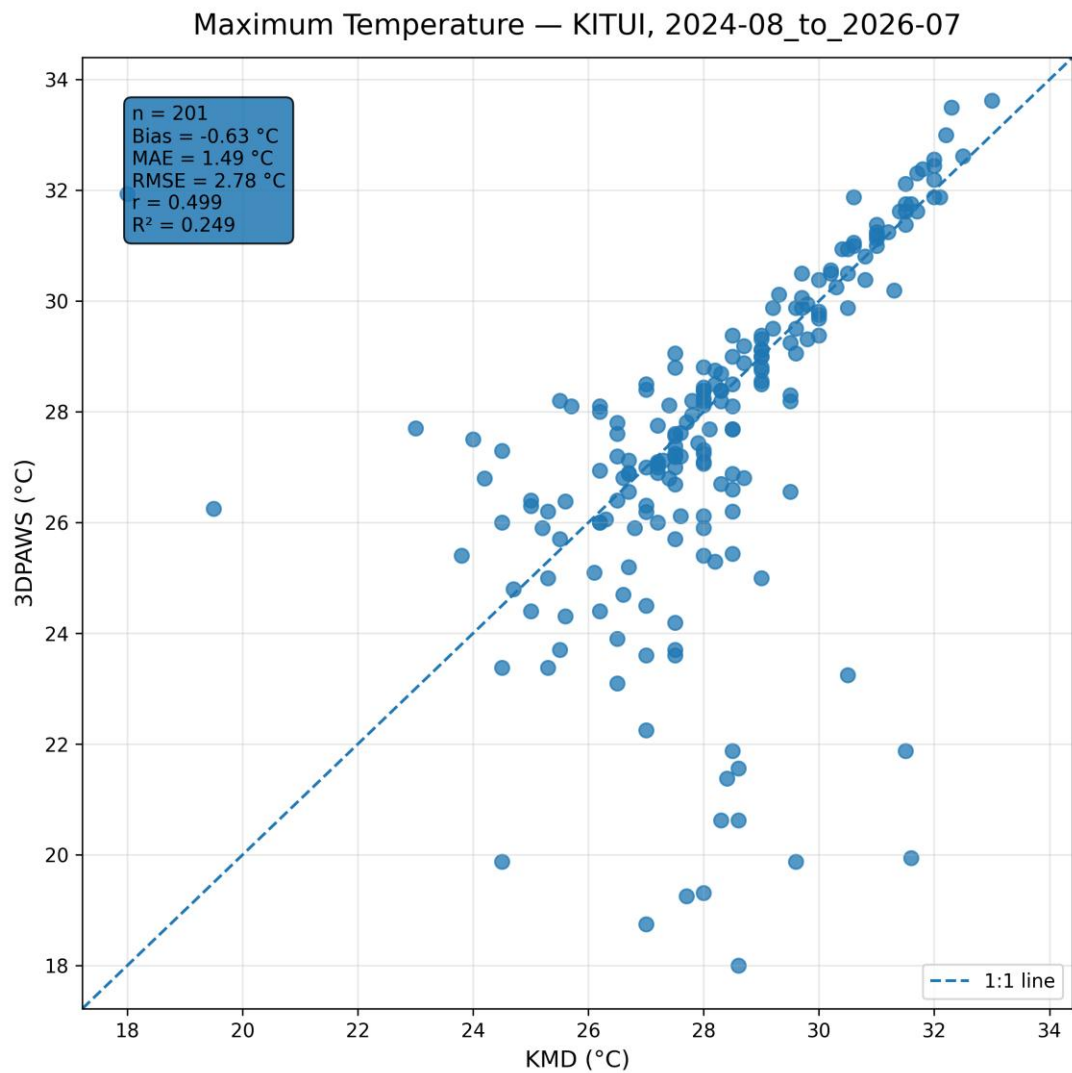


Figure 7:3 scatter comparison for Kitui — representative temperature validation result.

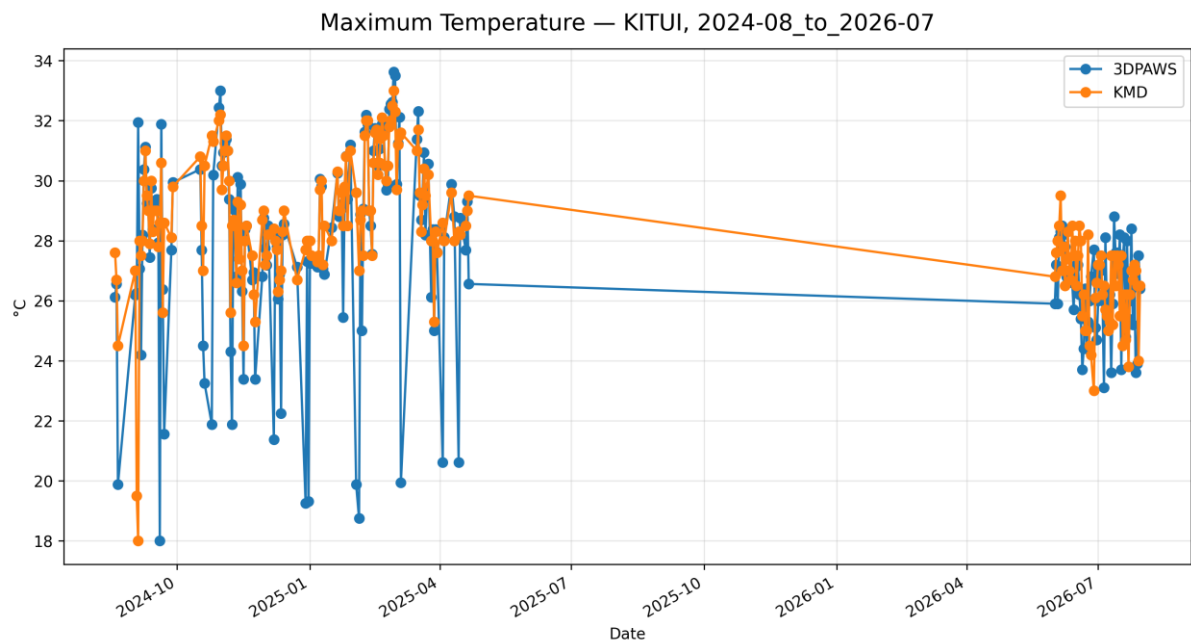


Figure 7:4 time-series comparison for Kitui — representative temperature validation result.

7.6 Relative Humidity Results

Relative humidity displayed a recurring pattern of strong temporal agreement but larger absolute differences. A high r value indicates that the 3D-PAWS system follows changes in humidity over time, but it does not by itself indicate small absolute errors. Several stations therefore achieved high correlation while remaining Below Threshold under the implemented humidity criteria.

Table 7:8 Cross-station humidity performance highlights

Station	Highest-r parameter	Highest r	Lowest-RMSE parameter	Lowest RMSE	Assessment	Data Status
Eldoret	Mean Relative Humidity	0.912	Mean Relative Humidity	5.424	Below Threshold	High Quality
Garissa	Mean Relative Humidity	0.888	Minimum Relative Humidity	12.235	Below Threshold	High Quality
Kitale - New	Mean Relative Humidity	0.833	Mean Relative Humidity	8.701	Below Threshold	High Quality
Kitui	Not assessed	—	—	—	—	—
Msabaha Agromet Station	Mean Relative Humidity	0.814	Minimum Relative Humidity	6.483	Below Threshold	Limited Quality
Mtwapa Agromet Station	Mean Relative Humidity	0.714	Minimum Relative Humidity	5.994	Below Threshold	High Quality
Narok	Mean Relative Humidity	0.891	Mean Relative Humidity	5.805	Below Threshold	High Quality
Nyahururu Agromet Station	Mean Relative Humidity	0.923	Minimum Relative Humidity	14.128	Below Threshold	High Quality

Eldoret illustrates strong temporal agreement in mean relative humidity, while Nyahururu illustrates strong temporal association for minimum humidity despite a substantial absolute difference.

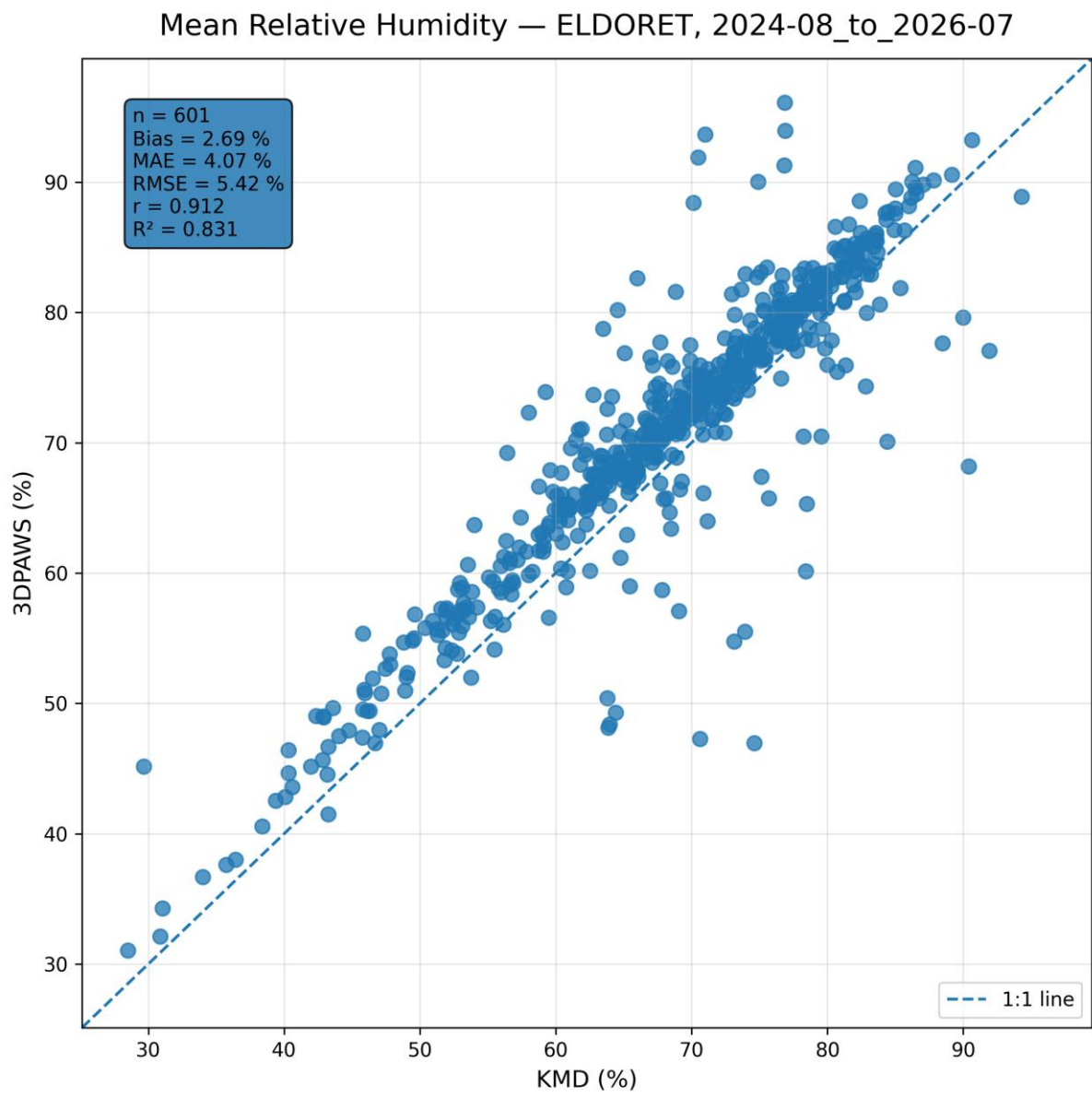


Figure 7:5 scatter comparison for Eldoret — representative humidity validation result.

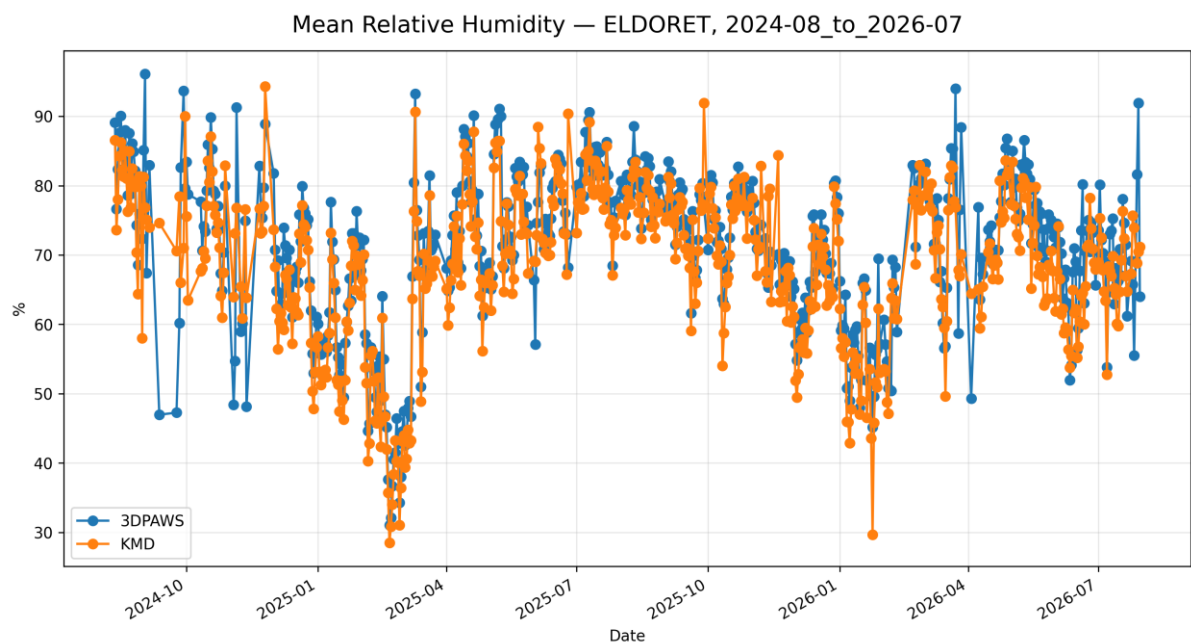


Figure 7:6 time-series comparison for Eldoret — representative humidity validation result.

7.7 Pressure Results

Atmospheric pressure was the most problematic parameter group overall. Pressure results varied considerably between stations and between maximum, minimum and mean pressure. The reports repeatedly identify elevation correction, pressure-reduction procedures, sensor response and station exposure as factors requiring further investigation.

Table 7:9 Cross-station pressure performance highlights

Station	Highest-r parameter	Highest r	Lowest-RMSE parameter	Lowest RMSE	Assessment	Data Status
Eldoret	Maximum Pressure	0.332	Mean Pressure	2.229	Below Threshold	High Quality
Garissa	Mean Pressure	0.678	Mean Pressure	6.420	Below Threshold	Good Quality
Kitale - New	Minimum Pressure	0.570	Mean Pressure	5.561	Below Threshold	High Quality
Kitui	Not assessed	—	—	—	—	—
Msabaha Agromet Station	Maximum Pressure	0.285	Maximum Pressure	10.456	Below Threshold	Limited Quality
Mtwapa Agromet Station	Minimum Pressure	0.924	Minimum Pressure	43.380	Below Threshold	Acceptable Quality
Narok	Maximum Pressure	0.794	Maximum Pressure	2.916	Below Threshold	High Quality
Nyahururu Agromet Station	Not assessed	—	—	—	—	—

The Mtwapa maximum-pressure result is based on only five paired days in the source report; it is therefore presented as a data-limited case rather than as evidence of general pressure behaviour.

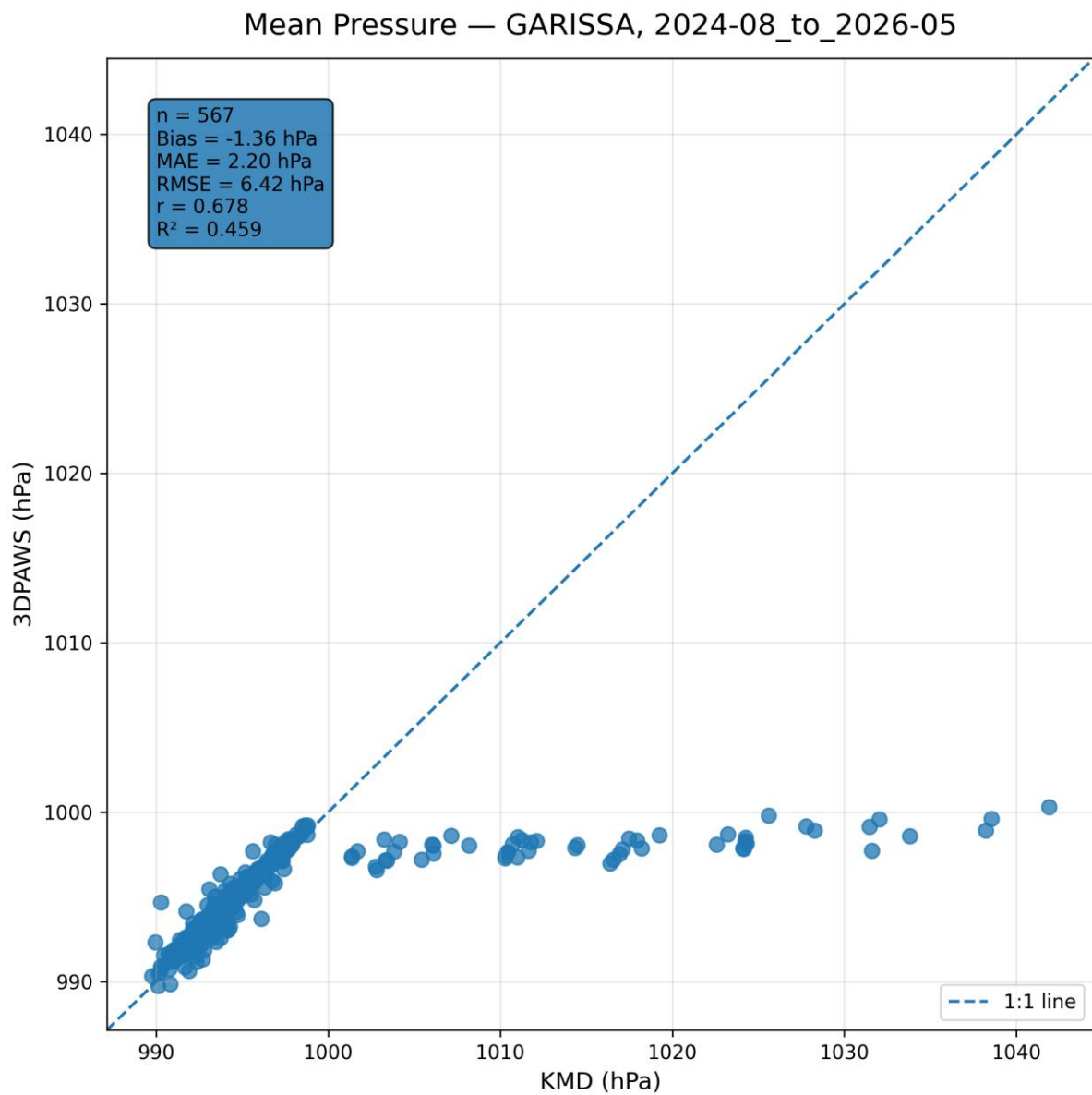


Figure 7:7 scatter comparison for Garissa — representative pressure validation result.

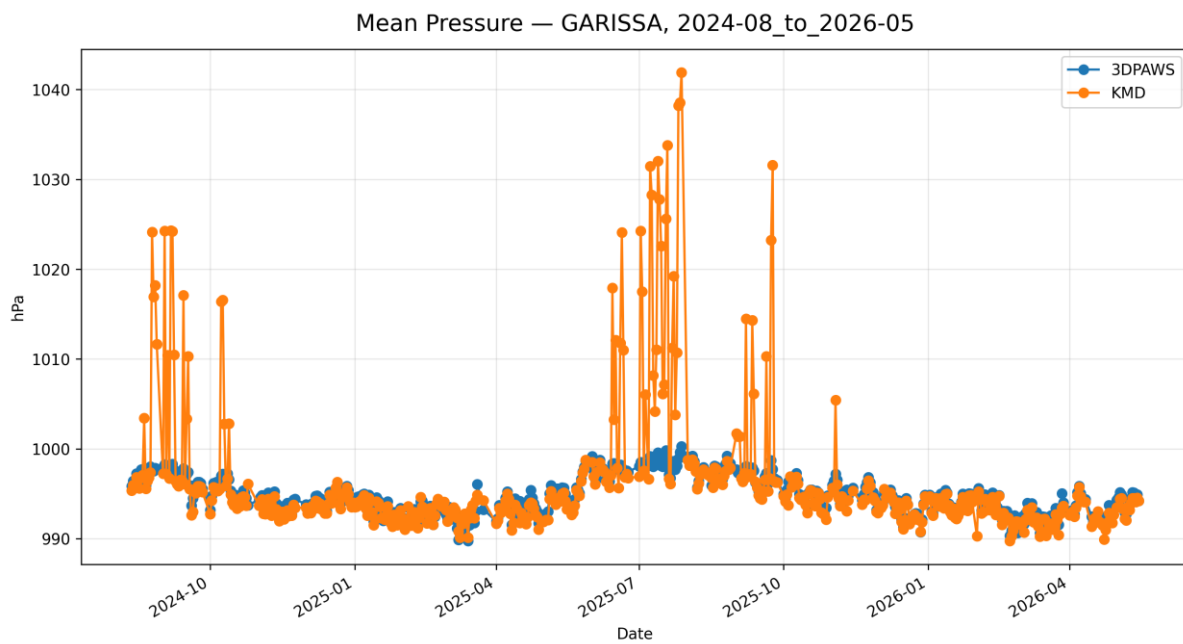


Figure 7:8 time-series comparison for Garissa — representative pressure validation result.

7.8 Rainfall Results

Rainfall results generally demonstrate the ability of the 3D-PAWS tipping-bucket system to capture rainfall occurrence and event variability. Differences in rainfall magnitude should be interpreted in relation to the event-based nature of precipitation, spatial variability between gauges and differences in gauge characteristics. The rainfall methodology also shows that agreement between the two 3D-PAWS rain gauges is an important diagnostic for identifying potential gauge malfunction.

Table 7:10 Cross-station rainfall performance highlights

Station	Highest-r parameter	Highest r	Lowest-RMSE parameter	Lowest RMSE	Assessment	Data Status
Eldoret	Daily Rainfall (Rain Gauge 1)	0.790	Daily Rainfall (Rain Gauge 1)	6.508	Below Threshold	Available
Garissa	Daily Rainfall (Rain Gauge 1)	0.536	Daily Rainfall (Rain Gauge 1)	5.885	Below Threshold	Available
Kitale - New	Daily Rainfall (Rain Gauge 2)	0.142	Daily Rainfall (Rain Gauge 2)	12.425	Below Threshold	Available
Kitui	Daily Rainfall (Rain Gauge 1)	0.636	Daily Rainfall (Rain Gauge 1)	9.850	Below Threshold	Available
Msabaha Agromet Station	Daily Rainfall (Rain Gauge 1)	0.500	Daily Rainfall (Rain Gauge 1)	8.237	Below Threshold	Available
Mtwapa Agromet Station	Daily Rainfall (Rain Gauge 1)	0.506	Daily Rainfall (Rain Gauge 2)	9.821	Below Threshold	Available
Narok	Daily Rainfall (Rain Gauge 1)	0.618	Daily Rainfall (Rain Gauge 1)	5.325	Below Threshold	Available
Nyahururu Agromet Station	Daily Rainfall (Rain Gauge 1)	0.697	Daily Rainfall (Rain Gauge 1)	7.606	Below Threshold	Available

Eldoret and Nyahururu provide a useful contrast between stronger and moderate rainfall temporal agreement.

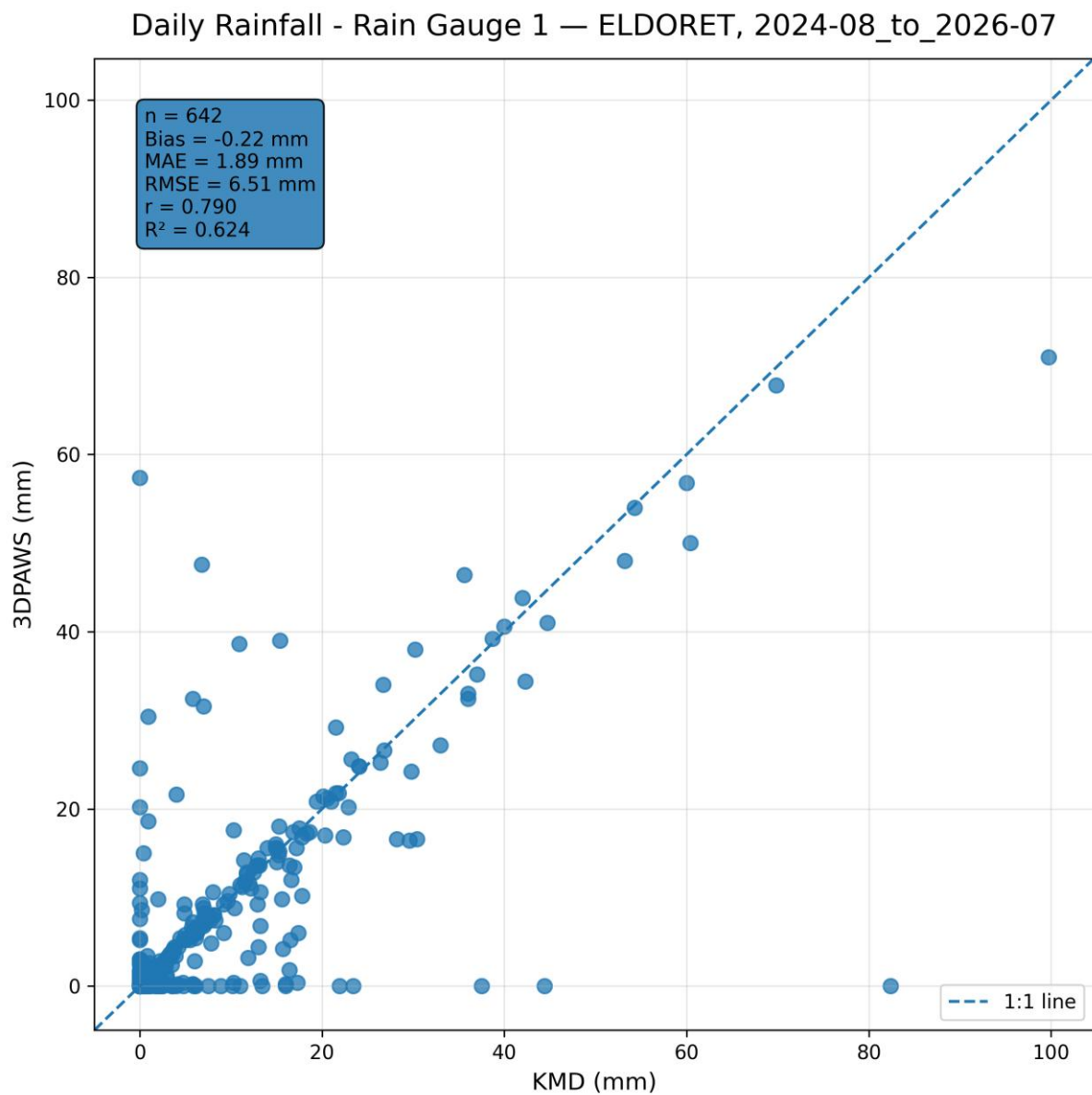


Figure 7:9 scatter comparison for Eldoret — representative rainfall validation result.

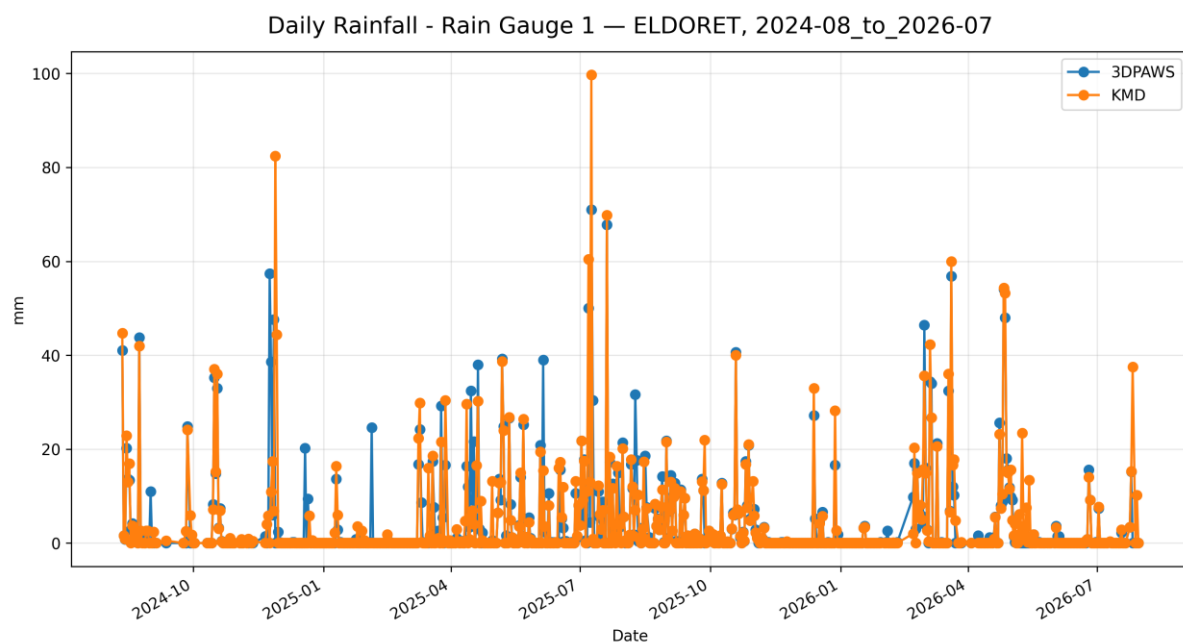


Figure 7:10 time-series comparison for Eldoret — representative rainfall validation result.

7.9 Cross-Station Comparison and Network-Level Interpretation

The cross-station results show substantial spatial and data-quality differences. Garissa stands out for strong temperature performance and high daily coverage. Narok shows a balanced pattern, including high temperature and humidity correlations and strong rainfall agreement. Eldoret also demonstrates strong temporal humidity agreement and useful temperature and rainfall performance. Msabaha provides strong temperature agreement but shows reduced humidity and pressure data quality. Mtwapa demonstrates strong temperature agreement but requires particular caution in interpreting pressure because of very limited paired coverage for some pressure statistics. Kitale-New is characterized by high sensor-level validity but low daily paired coverage, making its statistical estimates less representative of the full validation period. Kitui has useful rainfall coverage but limited-quality temperature, humidity and pressure data. Nyahururu shows good temperature and humidity temporal relationships but substantial absolute humidity differences.

A key finding across the network is that the classification Below Threshold does not necessarily mean that the station failed to reproduce meteorological variability. In several cases, particularly relative humidity and rainfall, the 3D-PAWS observations followed the temporal pattern of the KMD reference station while exhibiting systematic or magnitude differences. This distinction is central to interpreting the network as an observation-densification system rather than as a direct replacement for a fully established reference station.

Table 7:11 Station-level strongest agreement indicators

Station	Highest r parameter	Highest r	Smallest bias parameter	Bias	Lowest RMSE parameter	RMSE
Eldoret	Mean Relative Humidity	0.912	Maximum Temperature	-0.088	Maximum Temperature	1.306
Garissa	Maximum Temperature	0.940	Daily Rainfall (Rain Gauge 1)	-0.340	Maximum Temperature	0.776
Kitale - New	Mean Relative Humidity	0.833	Minimum Temperature	-0.202	Mean Temperature	2.085
Kitui	Daily Rainfall (Rain Gauge 1)	0.636	Daily Rainfall (Rain Gauge 1)	-0.054	Mean Temperature	2.129
Msabaha Agromet Station	Maximum Temperature	0.889	Daily Rainfall (Rain Gauge 1)	-0.093	Mean Temperature	0.889
Mtwapa Agromet Station	Minimum Pressure	0.924	Mean Temperature	-0.182	Mean Temperature	0.760
Narok	Minimum Temperature	0.892	Daily Rainfall (Rain Gauge 2)	-0.003	Minimum Temperature	1.277
Nyahururu Agromet Station	Mean Relative Humidity	0.923	Maximum Temperature	0.235	Mean Temperature	1.465

7.10 Overall Performance Assessment

Across the eight station reports, 79 parameter-level assessments are formally completed. The assessment distribution was 5 Breakthrough, 13 Threshold and 61 Below Threshold. The distribution reflects the strict parameter-specific criteria implemented in the validation toolkit and should not be interpreted as a simple pass/fail score for the entire 3D-PAWS network.

- a) **Temperature** was the most consistently favourable parameter group. Garissa recorded the strongest maximum-temperature agreement, with $r = 0.940$, $R^2 = 0.883$ and $RMSE = 0.776$ °C. Msabaha also showed strong maximum-temperature agreement ($r = 0.889$), while Mtwapa and Narok demonstrated strong correlations for mean and minimum temperature respectively.
- b) **Relative humidity** produced some of the strongest temporal correlations but also some of the largest absolute biases. Eldoret mean relative humidity achieved $r = 0.912$ and $R^2 = 0.831$, while Nyahururu minimum relative humidity achieved $r = 0.908$. Nevertheless, humidity classifications remained Below Threshold in these cases because absolute differences exceeded the implemented humidity criteria. This suggests that sensor calibration and exposure conditions require further attention before using these observations for applications that depend on high absolute humidity accuracy.
- c) **Pressure** was the least consistent parameter group. Some stations showed reasonable agreement in pressure magnitude, but temporal correlations are often weak and pressure statistics are affected by elevation correction and pressure-reduction procedures. The extreme Mtwapa maximum-pressure result is particularly data-limited because only five paired observations are available. Such results reinforce the need to assess pressure using sufficiently long, quality-controlled paired datasets before drawing operational conclusions.
- d) **Rainfall** showed useful temporal agreement at several stations, with Eldoret Rain Gauge 1 achieving $r = 0.790$ and Narok Rain Gauge 1 achieving $r = 0.618$. The rainfall results also demonstrate the importance of treating the two 3D-PAWS rain gauges as an internal quality-control pair. The supporting rainfall methodology reports near-perfect internal gauge correlations at Msabaha (1.00), Mtwapa (0.99) and Nyahururu (1.00), while Kitui showed a lower internal correlation of 0.68.

7.11 Key Findings

- The validation demonstrates that the locally fabricated 3D-PAWS stations can reproduce meaningful temporal variability in temperature, relative humidity and rainfall across different Kenyan environmental settings.
- Temperature is the strongest-performing parameter group overall. Garissa recorded the strongest maximum-temperature agreement, with $r = 0.940$, $R^2 = 0.883$ and $RMSE = 0.776$ °C.
- Relative humidity frequently showed strong temporal agreement but larger absolute differences. Eldoret mean relative humidity achieved $r = 0.912$ and $R^2 = 0.831$, demonstrating strong temporal reproduction despite a bias of 2.686%.
- Pressure requires the greatest additional investigation. The results point to elevation correction, pressure-reduction procedures, sensor response and station exposure as important factors requiring refinement and further validation.
- Rainfall observations generally captured rainfall occurrence and event variability. Differences in rainfall magnitude should be interpreted in relation to precipitation spatial variability and the characteristics of tipping-bucket gauges.
- Data completeness is a major determinant of the strength of the validation evidence. Kitale-New had daily coverage of only about 21.7–25.2%, while Garissa achieved 88.0–99.1%. Therefore, station-level statistics should always be reported with paired-observation coverage.
- The dual-rain-gauge configuration provides an important internal quality-control mechanism. Strong agreement between gauges can support confidence in rainfall event detection, while poor gauge-to-gauge agreement can flag potential mechanical, calibration or installation problems.

- The network results support the use of 3D-PAWS for observation-network densification and local meteorological monitoring, while also identifying clear priorities for calibration, quality control, pressure correction and continued comparative evaluation.

7.12 Interpretation for Operational Deployment

The results indicate that 3D-PAWS should be viewed as a complementary observation technology for network densification, particularly where increasing spatial coverage is more important than achieving reference-station-level accuracy for every parameter. The evidence is strongest for temperature and rainfall occurrence, while relative humidity shows useful temporal information but requires further calibration for absolute agreement. Pressure requires additional technical work before it can be treated with the same level of confidence.

The results also demonstrate the value of the integrated validation and visualization system developed for the study. The toolkit does not only generate performance statistics; it makes it possible to identify data gaps, inspect paired observations, examine scatter relationships and time-series behaviour, and distinguish persistent differences from isolated events. This provides an evidence base for targeted maintenance and calibration rather than relying on a single overall station score.

Before wider operational deployment, the results support continued field calibration, refinement of pressure-reduction and elevation-correction procedures, strengthening of automated quality-control rules, investigation of sensor exposure and siting, and continued comparison with KMD reference observations over longer periods. Stations with low paired-day coverage should be re-evaluated after additional observations have accumulated.

7.13 Chapter Summary

Overall, the validation assessment provides evidence that the 3D-PAWS platform can generate meteorological observations that reproduce important temporal variability across a geographically diverse set of Kenyan stations. The strongest results are observed in temperature, with useful rainfall agreement at several stations and strong temporal humidity agreement in selected cases. The main limitations are parameter-specific absolute differences, pressure behaviour, data completeness at some stations and the need for continued quality-control refinement. These results support the broader objective of using locally fabricated 3D-PAWS systems to densify meteorological observation networks while maintaining an evidence-based pathway toward improved calibration and operational readiness.

Source note: All station statistics, classifications, coverage values and representative figures in this consolidated chapter are derived from the uploaded station-level Chapter 7 validation reports generated using the 3D-PAWS Validation Toolkit. The chapter preserves the reported values and explicitly flags cases where coverage is limited.

8 DISCUSSION, CONCLUSIONS AND RECOMMENDATIONS

8.1 Introduction

This chapter discusses the principal findings of the multi-station validation of 3D-Printed Automatic Weather Stations (3D-PAWS) against Kenya Meteorological Department (KMD) reference observations and draws conclusions from the evidence presented in Chapter 7. The discussion focuses on the practical meaning of the statistical results rather than repeating the full numerical tables. Particular attention is given to the distinction between temporal agreement and absolute measurement accuracy, the influence of data completeness and quality, parameter-specific performance, rainfall-gauge consistency, and the implications of the findings for meteorological observation-network densification.

The validation covered eight 3D-PAWS/KMD station pairs and 79 formal parameter-level assessments across contrasting Kenyan geographical and agro-environmental settings. The assessment distribution comprised 5 Breakthrough, 13 Threshold and 61 Below Threshold classifications under the criteria implemented in the validation toolkit. These classifications provide an important performance reference; however, they should not be interpreted as a single network-wide pass/fail score. Performance must instead be considered together with correlation, error magnitude, paired-observation coverage, data quality, station conditions and intended operational use.

8.2 Discussion of the Major Findings

8.2.1 Overall performance of 3D-PAWS

The results demonstrate that 3D-PAWS can reproduce meaningful meteorological variability across a geographically diverse set of Kenyan stations. The strongest evidence is found in temperature, where several stations achieved Threshold or Breakthrough classifications and high correlations are obtained at stations such as Garissa, Msabaha, Mtwapa, Narok and Nyahururu. This indicates that the locally fabricated systems are capable of tracking important temporal changes in near-surface meteorological conditions.

At the same time, the results do not support treating all 3D-PAWS parameters as equivalent in performance. Relative humidity, pressure and rainfall exhibited parameter-specific limitations. The network therefore demonstrates the feasibility of 3D-PAWS as a complementary observation platform, but deployment should be accompanied by parameter-specific calibration, quality control, maintenance and continued comparison with KMD reference observations.

The multi-station design is a significant strength because it demonstrates both successful cases and cases requiring further technical improvement. The observed differences between stations indicate that performance is influenced not only by sensor characteristics, but also by environmental conditions, station exposure, data availability and station-specific processing.

8.2.2 Temperature performance

Temperature was the most consistently favorable parameter group. Garissa maximum temperature recorded $r = 0.940$, $R^2 = 0.883$ and $RMSE = 0.776$ °C, providing the strongest example of close temporal and numerical agreement. Results from other stations similarly indicate that the temperature sensors can reproduce daily and seasonal variability with useful consistency.

The comparatively weaker temperature results at stations such as Kitui and Kitale-New should be interpreted together with their data-quality and coverage limitations. In particular, Kitale-New had substantially lower paired-day coverage than stations with stronger records. Consequently, differences between stations cannot be attributed to sensor performance alone; record completeness and the number of valid paired observations also influence the reliability of the statistics.

The temperature findings therefore provide the strongest evidence for the immediate usefulness of 3D-PAWS in applications where temperature monitoring and spatial densification are priorities.

8.2.3 Relative humidity: temporal agreement versus absolute accuracy

Relative humidity produced one of the most important findings of the study. Several stations exhibited strong temporal relationships with KMD reference observations while simultaneously showing relatively large absolute differences. Eldoret mean relative humidity, for example, recorded $r = 0.912$ and $R^2 = 0.831$, demonstrating that the 3D-PAWS system closely followed the temporal evolution of humidity. However, its bias and error magnitude resulted in a Below Threshold classification.

This pattern demonstrates why correlation alone is insufficient for evaluating sensor performance. A sensor can reproduce the timing and direction of changes very well while remaining systematically offset from the reference. For operational applications requiring accurate absolute humidity values, calibration, sensor exposure, radiation protection, ventilation and installation conditions remain important considerations.

The humidity findings therefore suggest that the 3D-PAWS systems are potentially useful for monitoring temporal humidity variability, but further calibration and field evaluation are required before assuming reference-level absolute accuracy across all stations.

8.2.4 Atmospheric pressure and elevation correction

Pressure was the least consistent parameter group in the validation network. The results included weak temporal correlations and, at some stations, substantial differences in pressure magnitude. The evidence points toward pressure reduction, elevation correction, sensor response and station exposure as important factors requiring further investigation.

Pressure results should also be interpreted in the context of data availability. The Mtwapa maximum-pressure assessment, for example, was based on only five paired observations. Such a small sample is insufficient for robust generalization of long-term station behavior, even where the resulting statistical metrics appear large or unusual.

The findings indicate that pressure processing requires a dedicated refinement stage. Future comparison should distinguish raw station pressure from pressure reduced to a common reference level and verify the elevation and temperature assumptions used in the reduction procedure. Longer periods of quality-controlled paired observations are required before pressure performance can be conclusively characterized.

8.2.5 Rainfall performance and dual-gauge quality control

The rainfall results demonstrate useful capability for reproducing rainfall occurrence and event variability. Eldoret Rain Gauge 1 achieved $r = 0.790$, while Narok Rain Gauge 1 achieved $r = 0.618$. These results indicate that the 3D-PAWS rainfall observations contain meaningful temporal information when compared with KMD reference observations.

Rainfall is inherently more spatially variable than temperature or pressure, and the magnitude recorded by nearby gauges can differ because of rainfall intermittency, gauge exposure, wind effects, siting and mechanical characteristics. Consequently, rainfall disagreement should not automatically be interpreted as sensor failure.

The two-gauge configuration provides an additional quality-control opportunity. Strong agreement between the two gauges can increase confidence in rainfall event detection, while persistent disagreement can identify a gauge requiring inspection, cleaning, calibration or mechanical maintenance. The internal gauge comparison is therefore an important operational feature of the deployed system.

8.2.6 Importance of data completeness and observation coverage

Data completeness is a major determinant of the strength of the validation evidence. The network contained substantial differences in paired-day coverage. Garissa achieved coverage ranging from approximately 88% to 99%, whereas Kitale-New had coverage of approximately 22% to 25%. Mtwapa also contained pressure results with extremely limited paired coverage.

These differences demonstrate that statistical performance must always be reported together with the amount of paired data supporting the assessment. A high correlation calculated from a short or highly selective record should not be interpreted with the same confidence as a comparable correlation supported by a long, continuous record.

Future validation cycles should therefore establish minimum paired-data requirements before a parameter is formally classified. This would improve comparability between stations and prevent data-limited assessments from having disproportionate influence on network-level conclusions.

8.2.7 Spatial and agro-environmental diversity

The eight stations provide representation of contrasting Kenyan geographical and agro-environmental settings, including the Western Trans-Nzoia high-rainfall zone, Central Nyandarua/Nyahururu high-altitude zone, South Rift Narok agro-pastoral zone, South East Makueni/Kitui marginal-agricultural zone, Eastern Marsabit/Garissa/Tana River pastoral zone, and Coastal Kilifi/Taita-Taveta coastal zone. Eldoret provides an additional Rift Valley representation.

The differences observed between stations demonstrate that 3D-PAWS performance is influenced not only by the sensor itself but also by environmental conditions, station exposure, local micro-environment, data availability and station-specific processing. The multi-station design therefore provides a broader evidence base than a single-site assessment.

8.2.8 Local fabrication, modularity and operational experience

The deployment also demonstrates the practical value of local fabrication and modular system design. Additive manufacturing of structural components and local station assembly provide flexibility for customization, repair and incremental upgrading. The modular approach can facilitate replacement of damaged components without requiring complete station replacement and can contribute to development of local technical capacity.

These practical benefits are important where conventional automatic weather stations may be costly or logistically difficult to deploy and maintain. Affordability and scalability, however, do not remove the need for calibration, maintenance, quality control and reference-based validation.

8.3 Implications for Meteorological Observation Network Densification

The results support the potential of locally fabricated 3D-PAWS systems as a complementary means of increasing the spatial density of meteorological observations. The strongest case is not that every 3D-PAWS sensor currently matches a conventional reference instrument for every parameter, but that the platform can provide useful meteorological information at greater spatial coverage when supported by appropriate calibration, quality control and maintenance.

3D-PAWS can fill spatial gaps, support local monitoring and provide additional observations between conventional stations. Reference stations remain important for calibration, benchmarking and long-term climate-quality observations, while a denser 3D-PAWS network can provide additional spatial information.

The affordability and modularity of the system are particularly relevant to arid and semi-arid lands, agricultural regions and other underserved areas. The findings therefore support gradual expansion while retaining a representative subset of stations for intensive, long-term comparison with KMD reference stations.

The integration of data visualization with statistical assessment further strengthens the operational case. A system that continuously displays observations, detects anomalous behavior and provides comparative statistics can support more efficient maintenance, quality-control decisions and station prioritization.

8.3.1 Relevance to Regional Association I and developing-country observing networks

The Kenyan experience demonstrates that locally assembled, modular, low-cost systems can complement existing observation infrastructure and contribute to increasing observation density. The combination of additive manufacturing, commercially available sensors, solar-powered operation and cellular communication provides a practical development pathway that may be adapted by other National Meteorological and Hydrological Services where appropriate. Transferability nevertheless depends on local environmental conditions, technical capacity, calibration infrastructure and application-specific accuracy requirements.

8.3.2 Potential integration with WIS2 and OSCAR/Surface

Although the current implementation uses the CHORDS platform and a custom monitoring dashboard, future interoperability with the WMO Information System 2.0 (WIS2) could support standardized data exchange and dissemination. Documenting station metadata using the OSCAR/Surface framework could improve discoverability, metadata management and international interoperability. Such integration would strengthen the contribution of low-cost observations to national, regional and global observing systems, provided that data-quality and metadata requirements are satisfied.

8.4 Conclusions

Based on the multi-station validation evidence presented in Chapter 7, the following conclusions are drawn:

- Locally fabricated 3D-PAWS systems are capable of producing meteorological observations that reproduce meaningful temporal variability when compared with KMD reference observations.
- Temperature is the strongest-performing parameter group. The Garissa maximum-temperature result, with $r = 0.940$, $R^2 = 0.883$ and $RMSE = 0.776$ °C, provides strong evidence of the system's capability to reproduce temperature variability with relatively small error.
- Relative humidity demonstrates strong temporal reproduction at several stations, but relatively large absolute differences at some sites show that temporal correlation does not necessarily imply reference-level accuracy.
- Atmospheric pressure remains the principal technical area requiring further refinement. Pressure reduction, elevation correction, sensor response, exposure and data completeness require additional investigation.
- The 3D-PAWS rainfall system provides useful information on rainfall occurrence and temporal variability. The dual-rain-gauge arrangement provides an additional internal consistency check for rainfall quality control.
- The quality and completeness of paired datasets substantially influence the strength and reliability of validation statistics. Future assessments should enforce minimum paired-data requirements and retain coverage alongside performance metrics.
- The multi-station results support the potential of 3D-PAWS for observation-network densification, particularly as a complementary system rather than an immediate universal replacement for conventional reference stations.
- The integrated validation toolkit provides an evidence-based framework combining data-quality assessment, statistical validation, graphical comparison and performance classification to guide station improvement and operational readiness.

8.5 Recommendations

8.5.1 Sensor calibration and field verification

- Establish a structured calibration programme for temperature, relative humidity, pressure and rainfall sensors before deployment and at defined intervals during operation.
- Use KMD reference observations as the principal field benchmark for continued verification and recalibration.
- Record calibration history, sensor replacement, maintenance and environmental exposure information for each station.

8.5.2 Pressure processing and elevation correction

- Undertake a dedicated review of the pressure-reduction algorithm, station elevations and reference-pressure conventions used by the validation toolkit.
- Separate raw station-pressure validation from reduced-pressure validation so that sensor behavior and reduction-method errors can be distinguished.
- Repeat pressure validation using longer, quality-controlled paired records before making operational decisions based on pressure performance.

8.5.3 Data quality control

- Strengthen automated quality-control procedures to identify missing data, persistent offsets, unrealistic values, abrupt changes, sensor drift and inconsistent rainfall-gauge behaviour.
- Retain parameter-specific quality flags and ensure that network-level statistics remain traceable to the original station and validation period.
- Introduce explicit minimum paired-observation thresholds before calculating or classifying validation statistics.

8.5.4 Rainfall gauge management

- Use the two 3D-PAWS rainfall gauges as an internal consistency check and investigate persistent gauge-to-gauge disagreement.
- Maintain regular inspection and maintenance schedules for tipping-bucket gauges and verify mechanical operation after extreme rainfall events or prolonged dry periods.
- Interpret rainfall differences with consideration of local siting, wind exposure and precipitation spatial variability.

8.5.5 Network expansion

- Expand the validation network gradually, prioritizing areas with significant spatial observation gaps and contrasting environmental conditions.
- Retain a representative subset of stations for intensive long-term comparison with KMD reference stations.
- Do not use the current network-wide assessment as the sole basis for declaring operational equivalence; continue collecting evidence as additional observations become available.

8.5.6 Regular monitoring and maintenance

- Integrate automated station-health monitoring with the visualization platform so that battery, communications, data continuity and sensor anomalies can be identified early.
- Develop station-specific maintenance records and performance histories to support targeted interventions.
- Use validation results to prioritize stations or parameters requiring recalibration rather than applying the same intervention uniformly across the network.

8.6 Areas for Further Research and Development

The present validation provides a strong basis for continued development and deployment of the 3D-PAWS network; however, several areas require further investigation before the system can be considered fully mature for wider operational use. Future work should therefore focus on strengthening the validation evidence, understanding the influence of environmental and site-specific conditions on sensor performance, improving automated quality control and anomaly detection, and enhancing the supporting data and monitoring infrastructure. Particular attention is required for stations and parameters with limited paired-data coverage, while seasonal analysis and assessment of station exposure and micro-environmental effects will help distinguish sensor-related limitations from genuine spatial differences between 3D-PAWS and KMD observations. Further development should also consider additional statistical and event-based evaluation approaches, particularly for rainfall, alongside continued improvements to data storage, visualization and network-health monitoring. As the network expands, research should evaluate the value of the additional observations for local forecasting support, agricultural advisories and monitoring of spatial weather variability while maintaining application-specific accuracy requirements. Finally, continued national deployment should prioritize ASALs, agricultural regions and other underserved areas, with structured validation against KMD reference stations maintained throughout the expansion process.

The following areas are therefore recommended for further research and development:

- Extend validation periods for stations with low paired-data coverage, particularly Kitale-New and pressure observations at Mtwapa, to provide a more robust assessment of long-term performance.
- Investigate seasonal performance by separating wet- and dry-season observations and determining whether sensor biases vary systematically with changing environmental conditions.
- Investigate station exposure and micro-environmental effects to distinguish sensor-related limitations from genuine spatial differences between the 3D-PAWS and KMD observation sites.
- Develop automated anomaly-detection and quality-control methods capable of identifying sensor drift, anomalous observations and persistent bias in near-real time.
- Evaluate additional statistical and event-based measures, where appropriate, particularly for rainfall, where occurrence, intensity and accumulation characteristics may require additional evaluation beyond conventional continuous-observation statistics.
- Continue development of integrated data storage, visualization and monitoring infrastructure to support long-term operational use, data accessibility, network-health monitoring and maintenance.
- Evaluate the additional observations generated by a denser 3D-PAWS network for local forecasting support, agricultural advisories and monitoring of spatial weather variability, while respecting the accuracy requirements of each application.
- Continue expansion of the national 3D-PAWS network, prioritizing ASALs, agricultural regions and other underserved areas, while maintaining structured validation against KMD reference stations throughout the expansion process.

8.6.1 3D-PAWS Observation and Analysis Dashboard

A custom 3D-PAWS Observation and Analysis Dashboard is under development to support operational deployment of the national network. The dashboard is intended to provide real-time visualization, automated quality control, climatological summaries, station-health monitoring and historical data analysis directly from the CHORDS platform. The present study used exported CHORDS datasets and Python-based analysis; future versions will automate these workflows and provide an integrated operational decision-support platform.

8.6.2 AI-Based Quality Control

Future work will investigate Artificial Intelligence and machine-learning techniques for automated quality control, including anomaly detection, sensor-drift identification, equipment-fault diagnosis and predictive maintenance.

8.6.3 InfluxDB and Grafana Integration

Future development will investigate InfluxDB as a high-performance time-series database together with Advanced Grafana for advanced visualization, real-time analytics, customizable dashboards and long-term storage. This would complement CHORDS and strengthen operational monitoring and decision support.

8.6.4 WMO Digital Infrastructure Integration

Future work will explore interoperability with WIS2 and OSCAR/Surface to support standardized data exchange, metadata management and integration with national, regional and global meteorological observing systems.

8.7 Contribution of the Study

- a) The study contributes an applied validation framework for locally fabricated automatic weather stations under Kenyan field conditions. Its contribution extends beyond fabrication of individual stations by combining station deployment, sensor observation, data transmission, data-quality assessment, comparative statistics, graphical validation and network-level interpretation.
- b) The eight-station assessment demonstrates a practical method for identifying where 3D-PAWS already provides useful observational information and where further technical development is required. In particular, the study establishes an evidence base for continued improvement of temperature and rainfall observations, targeted refinement of relative-humidity performance, and dedicated investigation of pressure processing and correction.
- c) The resulting approach can support future expansion of locally fabricated observation networks while maintaining transparent comparison with established KMD reference observations. The combination of local fabrication, station-level monitoring and a reproducible validation workflow provides a practical pathway from prototype deployment toward evidence-based operational network densification.

8.8 Conclusion

The evidence generated in this study supports the conclusion that 3D-PAWS is a promising and practical technology for increasing the density of meteorological observations in Kenya. Across eight stations, the systems demonstrated the ability to reproduce important temporal variability, particularly for temperature and, at several locations, rainfall and relative humidity. However, the results also show that performance is parameter-specific and strongly influenced by calibration, station conditions, data completeness and processing procedures.

The most appropriate pathway is therefore continued deployment alongside structured validation rather than immediate replacement of conventional reference observations. With improved calibration, pressure-reduction procedures, automated quality control, systematic maintenance and longer validation records, 3D-PAWS can provide an increasingly valuable complementary layer within Kenya's meteorological observation network.

In this context, the study demonstrates not only the feasibility of locally fabricating automatic weather stations, but also a practical pathway for moving from prototype deployment toward evidence-based operational network densification. The principal value of the platform lies in combining affordability and scalability with transparent validation, continuous quality control and progressive technical improvement.

9.1 Deployment Costs

The 3D-Printed Automatic Weather Station (3D-PAWS) was developed with the objective of providing a cost-effective alternative for expanding meteorological observation networks, particularly in regions where deployment of conventional Automatic Weather Stations (AWS) is economically challenging. The use of locally fabricated 3D-printed components, commercially available sensors, open-source electronics, and in-house assembly significantly reduces the overall deployment cost.

In addition to lower procurement costs, the modular design minimizes maintenance expenses by enabling individual components to be repaired or replaced without requiring complete system replacement. Local fabrication also reduces dependence on imported structural components and shortens maintenance turnaround times.

Although deployment costs vary depending on communication infrastructure, power systems, and sensor configuration, the 3D-PAWS approach offers substantial savings while maintaining the capability to collect key meteorological observations.

9.2 Comparison with Conventional Automatic Weather Stations

Conventional Automatic Weather Stations are internationally recognized for providing high-accuracy meteorological observations and remain the reference standard for operational forecasting and climate monitoring. However, their acquisition, installation, calibration, and maintenance requirements may limit widespread deployment, particularly where financial and technical resources are constrained.

By comparison, the 3D-PAWS system emphasizes affordability, modularity, and ease of deployment. The use of 3D-printed components and open-source hardware allows rapid assembly and simplifies maintenance. These characteristics make the system particularly suitable for increasing observation density and complementing existing national meteorological networks.

Table 9:1 General Comparison between Conventional AWS and 3D-PAWS

Feature	Conventional AWS	3D-PAWS
Initial acquisition cost	High	Low
Local fabrication	No	Yes
Installation complexity	High	Moderate
Maintenance	Specialized	Modular and locally manageable
Component replacement	Vendor-dependent	Individual modules replaceable
Scalability	Limited by cost	Highly scalable
Communication	Various	Cellular
Power Supply	Solar/Grid	Solar and Battery
Intended Role	Reference observations	Network densification and complementary observations

Table 9:2 Estimated cost components: The cost estimates are applicable only within KMD network. Logistics and other challenges may be applicable to other networks or sites

Cost Component	Conventional AWS (USD) EST	3DPAWS (USD) EST
Station hardware	6,000	500 (for material import)
Locally sourced Material	15	45
Installation	2,000	400
Annual maintenance	3,000	300
Sensor replacement	2,000	150
Communication	15	15
Total estimated cost	13,030	1410

9.3 Cost Effectiveness

The affordability and modular architecture of the 3D-PAWS system provide an opportunity to deploy a greater number of observation stations within a given budget, thereby increasing the spatial density of meteorological observations. This is particularly important in arid and semi-arid lands (ASALs), rural communities, agricultural areas and other underserved locations, where conventional automatic weather stations may be limited by relatively high acquisition, installation and maintenance costs.

In addition to stations operated by the national meteorological service, Kenya's meteorological observation environment includes weather stations installed and operated by private-sector organizations, agricultural institutions, research organizations, development partners and other stakeholders. These stations can provide valuable observations at locations that may not otherwise be adequately represented within the national observation network. However, differences in instrumentation, data formats, observation intervals, quality-control procedures, data management practices and accessibility can make it difficult to integrate observations from different networks into a common operational framework.

The deployment of 3D-PAWS therefore presents an opportunity not only for network densification but also for the development of a more distributed and collaborative observation ecosystem. Low-cost stations can be deployed by different institutions and organizations while observations are brought together through appropriate data-management, quality-control and visualization systems.

The combination of local fabrication, modular construction, simplified maintenance and cloud-based data management has the potential to reduce the cost of expanding and maintaining observation networks. The integration of CHORDS and the custom monitoring dashboard developed in this study further provides a mechanism for centralized visualization, station-health monitoring, data availability assessment and retrieval of observations from geographically distributed stations.

In this context, 3D-PAWS should not be regarded as a replacement for high-end Automatic Weather Stations or existing private-sector observation infrastructure. Instead, it can serve as a complementary technology for expanding observation coverage. Conventional AWS can continue to provide high-quality reference observations at strategically important locations, while 3D-PAWS can provide additional observations in areas where conventional systems may be economically or operationally difficult to deploy. Similarly, observations from suitable private-sector and institutional stations can contribute additional spatial coverage where their data quality and metadata are adequately documented.

A coordinated approach involving national meteorological stations, 3D-PAWS stations, private-sector stations, research and agricultural observation networks could substantially improve the spatial representation of meteorological conditions. Such integration would be particularly valuable for localized weather forecasting, agricultural advisory services, early warning, hydrometeorological applications and climate monitoring.

However, integration of observations from multiple providers requires appropriate metadata standards, quality-control procedures, station identification, data-sharing arrangements and interoperability mechanisms. The use of common standards and platforms can facilitate the transformation of heterogeneous observations into datasets that can be more readily accessed and evaluated.

Therefore, the potential value of 3D-PAWS extends beyond the deployment of individual low-cost stations. Its greater contribution may be as part of a multi-source meteorological observation network, in which low-cost community and institutional stations complement national and private-sector infrastructure. This approach can increase observation density, improve data availability and provide a more geographically representative basis for weather and climate services while retaining conventional AWS as important reference and anchor stations.

Examples of 3D-PAWS Installed by private entities:

- Kenya Turkana South Kainuk (IRC) Kainuk Secondary (53)
- Kenya Turkana North Lokitaung (IRC) (54)
- Kenya Turkana East Lokori (IRC) (55)
- Kenya Turkana West Kakuma (IRC) (56)
- Kenya Kiambu JKUAT IOT AWS - Conduiti@Empathy1 (61)
- Kenya Isiolo Merti Power Plant (62)
- Kenya Mandera Elwak Power Plant (63)
- Kenya Wajir Habasweni Power Plant (64)
- Kenya Tana River Sugar Company: BuraTana (65)
- Kenya VOI Mengo Primary School (66) KALRO
- Kericho Tea Research (67)

CHAPTER TEN

10 OVERALL SYNTHESIS AND ACHIEVEMENT OF THE STUDY OBJECTIVES

10.1 Introduction

This chapter provides the final synthesis of the study on the evaluation and deployment of 3D-Printed Automatic Weather Stations (3D-PAWS) in Kenya with integrated data visualization and monitoring systems. It brings together the evidence presented in the preceding chapters, with particular emphasis on the multi-station validation results in Chapter 7, the interpretation and recommendations in Chapter 8, and the economic assessment presented in Chapter 9. The purpose is not to repeat the detailed results, but to establish what the study demonstrates as a whole, identify the principal contribution of the work, and relate the findings to the practical objective of strengthening meteorological observation-network density.

The study adopted a multi-station approach involving eight 3D-PAWS and KMD reference-station pairs representing contrasting Kenyan geographical and agro-environmental settings. The assessment covered

- temperature,
- relative humidity,
- atmospheric
- pressure
- rainfall,

supported by data-quality assessment, statistical comparison, graphical analysis and network-level aggregation. This combination provides a basis for assessing not only whether individual stations can produce useful observations, but also whether locally fabricated systems have practical potential as a complementary component of a wider meteorological observation network.

10.2 Overall Synthesis of the Study Findings

Taken together, the study demonstrates that locally fabricated 3D-PAWS can generate meteorological observations with meaningful temporal variability under Kenyan field conditions. The validation results showed that performance varies substantially by parameter and station. Temperature provided the strongest and most consistent evidence of useful agreement with KMD observations, while relative humidity showed several cases of strong temporal agreement accompanied by larger absolute differences. Atmospheric pressure was the least consistent parameter group and requires further refinement of pressure reduction, elevation correction and station-specific processing. Rainfall observations provided useful information on rainfall occurrence and temporal variability, while the dual-gauge configuration provided an additional internal quality-control mechanism.

The results also demonstrate the importance of distinguishing temporal agreement from absolute accuracy. A high correlation indicates that a 3D-PAWS observation follows changes in the reference observation, but does not by itself establish that the magnitude of the measurements is sufficiently accurate. The combined use of bias, MAE, RMSE, correlation, data coverage and quality classification therefore provides a more appropriate basis for evaluating the operational potential of the stations.

The network-level results further show that data completeness is an important component of validation. Stations and parameters with limited paired observations require more cautious interpretation than those supported by long and continuous records. This finding supports the continued use of minimum data-coverage requirements and parameter-specific quality-control procedures in future validation cycles.

10.3 Synthesis of the Geographical and Agro-Environmental Representation

The multi-station design provides evidence from contrasting environments rather than from a single representative site. The evaluated network includes

- Western Trans-Nzoia high-rainfall zone,
- Central Nyandarua/Nyahururu high-altitude zone,
- South Rift Narok agro-pastoral zone,
- South East Makueni/Kitui marginal-agricultural zone,
- Eastern Marsabit/Garissa/Tana River pastoral zone,
- Coastal Kilifi/Taita-Taveta coastal zone,
- Rift Valley representation at Eldoret.

This geographical diversity is important because low-cost station performance cannot be assumed to be uniform across environmental settings. Differences in temperature, humidity, rainfall regime, elevation, exposure and station conditions may influence sensor behavior and data quality. The study therefore supports a deployment strategy based on continued site-specific validation rather than assuming that performance demonstrated at one station automatically applies to all environments.

10.4 Achievement of the Study Objectives

The study objectives are addressed through the combined activities of system development, deployment, data acquisition, validation, visualization and economic assessment. The achievement of each objective is summarized below.

10.4.1 Development and deployment of the 3D-PAWS system

The study achieved the system-development and deployment component by implementing locally fabricated 3D-PAWS stations incorporating meteorological sensors, power, communications and data transmission components. The deployment demonstrated that locally manufactured structural components can be combined with commercially available sensing and communication technologies to produce functional automatic weather-station platforms.

10.4.2 Data acquisition, transmission and visualization

The study established an operational data pathway through the CHORDS platform and associated monitoring and analysis tools. The resulting workflow enabled station observations to be accessed, processed, compared with KMD reference observations and presented through graphical and statistical outputs. This provided the foundation for repeatable station-level and network-level validation.

10.4.3 Evaluation of observational performance

The observational-performance objective was addressed through the eight-station validation exercise. The study evaluated temperature, relative humidity, pressure and rainfall using paired observations and multiple statistical measures. The results establish that 3D-PAWS can reproduce meaningful meteorological variability, while also identifying parameter-specific limitations requiring further calibration and technical refinement.

10.4.4 Assessment of Sustainability and Suitability for Network Densification

The study provides evidence supporting the use of 3D-PAWS as a complementary and sustainable technology for meteorological network densification. Its affordability, local fabrication potential, modular architecture, simplified maintenance and cloud-based monitoring provide opportunities to expand observation coverage, particularly in underserved, rural, arid and semi-arid areas where conventional station deployment may be constrained by cost and logistics. The results do not support positioning 3D-PAWS as a universal replacement for conventional reference stations. Instead, they support a layered observing-network model, in which conventional stations continue to provide reference and high-quality observations, while 3D-PAWS and

appropriately managed partner or private-sector stations increase spatial coverage. This approach provides a practical and potentially sustainable pathway for strengthening Kenya's meteorological observation network.

10.4.5 Assessment of economic feasibility

The cost analysis in Chapter 9 provides the economic component of the study by examining the resources required to fabricate and deploy the 3D-PAWS system and comparing these requirements with conventional automatic weather-station approaches. The economic findings complement the technical validation by showing that the potential value of 3D-PAWS is associated not only with measurement capability but also with the possibility of increasing observation density through locally maintainable and scalable systems.

10.5 Contribution of the Study

The principal contribution of the study is the development and field evaluation of a locally fabricated meteorological observation approach supported by a reproducible validation and monitoring workflow. Rather than treating fabrication, deployment, data transmission and validation as separate activities, the study connects them into a single pathway from station construction to evidence-based assessment.

The study contributes empirical evidence from multiple Kenyan environments, including areas with contrasting rainfall, elevation and agro-environmental conditions. This multi-station evidence is particularly valuable because it demonstrates both successful performance and areas where further technical improvement is needed.

A further contribution is the validation workflow developed around station-level quality assessment, statistical comparison, graphical analysis, WMO-oriented performance assessment and network aggregation. The workflow provides a repeatable basis for future validation as additional stations and longer datasets become available.

Finally, the study contributes to the broader discussion of locally developed and cost-conscious observing technologies for developing-country meteorological services. It demonstrates a practical approach in which local fabrication and digital monitoring can be combined with established reference observations rather than operating independently of national meteorological infrastructure.

10.6 Practical Implications for Kenya's Meteorological Observation Network

The findings suggest that 3D-PAWS can be considered as one component of a broader strategy for increasing the density of meteorological observations in Kenya. Priority may be given to;

1. Spatial gaps,
2. Agricultural production areas,
3. Arid and semi-arid lands and
4. Other locations where conventional observing infrastructure is limited.

Deployment should, however, be accompanied by structured calibration, station metadata management, quality control and maintenance.

A representative subset of 3D-PAWS stations should continue to operate alongside KMD reference stations for long-term comparison. This would allow systematic identification of sensor drift, site effects and seasonal performance changes while providing an evidence base for determining which parameters and applications are appropriate for different levels of 3D-PAWS deployment.

The digital monitoring and validation infrastructure can further support operational management by providing station-health information, historical performance records and automated identification of anomalous observations. Integration with national and WMO digital information systems can be considered as the network matures and once data-quality and metadata requirements are satisfied.

10.7 Overall Research Limitations

The study should be interpreted within the limitations of the available datasets and validation periods. The eight stations did not all have identical validation periods or equal data completeness, and some parameters are supported by substantially fewer paired observations than others. These differences limit the extent to which network-level statistics can be interpreted as fully representative of all Kenyan environmental conditions.

In addition, rainfall measurements are intrinsically affected by spatial variability and gauge exposure, while pressure measurements are sensitive to elevation and reduction procedures. The results therefore cannot be generalized to all possible station configurations without additional testing. These limitations do not invalidate the study; rather, they define the conditions under which the conclusions should be applied and identify priorities for subsequent validation.

10.8 Overall Recommendations

Based on the findings of the study, the following recommendations are proposed to support the continued development, operationalization and national expansion of the 3D-PAWS network:

1. Continue expansion of 3D-PAWS in a controlled and evidence-based manner, prioritizing areas with significant observation gaps, particularly ASALs, agricultural regions and other underserved areas.
2. Maintain structured comparison between 3D-PAWS and KMD reference stations to support long-term calibration, performance monitoring and assessment of sensor stability.
3. Strengthen parameter-specific calibration, particularly for relative humidity and atmospheric pressure, where the validation results indicate greater differences from the reference observations.
4. Refine pressure-reduction and elevation-correction procedures and validate the resulting methods using longer paired datasets and observations from stations at different elevations.
5. Maintain the dual-rain-gauge configuration as an internal rainfall quality-control mechanism and investigate persistent disagreement between the two gauges to identify possible sensor, installation or exposure-related issues.
6. Introduce minimum paired-observation requirements before assigning formal statistical performance classifications, particularly where data coverage is insufficient to support robust conclusions.
7. Establish and implement a structured preventive maintenance schedule for all deployed stations. Routine maintenance should include inspection and cleaning of sensors, rain gauges and radiation shields; checking of connectors, cables and mounting structures; verification of battery and power-system performance; inspection of solar panels; assessment of cellular communication and data transmission; and confirmation of sensor operation and data availability. Maintenance activities should be documented in station-specific maintenance records, with corrective actions and replacement of faulty components tracked to completion.
8. Strengthen automated quality control, station-health monitoring and maintenance records to enable early identification of sensor drift, communication failures, power problems, missing observations and other operational faults.
9. Continue development of the integrated 3D-PAWS Observation and Analysis Dashboard, including station administration, zonal monitoring, data availability assessment, downloadable daily and monthly summaries, network-health indicators and future AI-based quality-control capability.
10. Investigate integration with InfluxDB and WMO digital infrastructure, including WIS2 and OSCAR/Surface, as the operational system matures and the network expands.
11. Retain a representative subset of conventional KMD reference stations for benchmarking and long-term validation, ensuring that the performance of the expanding 3D-PAWS network continues to be assessed against established reference observations.

10.9 Final Synthesis

The combined evidence from the technical development, multi-station validation and cost analysis demonstrates that 3D-PAWS has meaningful potential as a locally developed and scalable complementary technology for meteorological observation in Kenya. The study does not establish that all 3D-PAWS parameters currently achieve the accuracy of conventional reference systems under all conditions. Instead, it establishes a more useful and defensible conclusion: 3D-PAWS can provide valuable additional observations when deployed within a structured framework of calibration, quality control, maintenance, reference comparison and progressive technical improvement.

The strongest pathway emerging from the study is therefore a complementary network model. Conventional KMD stations provide high-quality reference observations and long-term continuity, while locally fabricated 3D-PAWS stations can increase spatial density and provide additional observations in underserved locations. With continued validation, improved sensor processing, stronger quality control and systematic maintenance, the platform can progress from a field-deployed prototype approach toward a more mature operational component of Kenya's meteorological observation infrastructure.

Overall, the study demonstrates that local fabrication, digital connectivity, systematic validation and cost-conscious network design can be combined to address observational-density challenges. The work therefore provides both an empirical assessment of 3D-PAWS performance under Kenyan conditions and a practical framework for continued development and responsible scaling of the technology.

11 List of abbreviations and definition of terms

Abbreviation	Full Term
3D-PAWS	3D-Printed Automatic Weather Station
API	Application Programming Interface
ASALs	Arid and Semi-Arid Lands
AWS	Automatic Weather Station
CHORDS	Cloud-Hosted Real-time Data Services
CSV	Comma-Separated Values
GPS	Global Positioning System
GSM	Global System for Mobile Communications
ICT	Information and Communication Technology
KMD	Kenya Meteorological Department
LTE	Long-Term Evolution
LoRa	Long Range
OSCAR	Observing Systems Capability Analysis and Review Tool
QC	Quality Control
RA I	Regional Association I
RH	Relative Humidity
WIS2	WMO Information System 2.0
WMO	World Meteorological Organization
UCSB	The University of California, Santa Barbara
UCAR	The University Corporation for Atmospheric Research
FEWS NET	The Famine Early Warning Systems Network
EMNP	Enhancing Meteorological Networks Partnership

LIST OF SYMBOLS AND STATISTICAL TERMS

Symbol	Term	Definition
r	Correlation coefficient	
R ²	Coefficient of determination	
MAE	Mean Absolute Error	
RMSE	Root Mean Square Error	
°C	Degrees Celsius	
hPa	Hectopascal	
mm	Millimetres	
%	Percentage	
n	Number of paired observations	
P	Atmospheric pressure	
T	Air temperature	
RH	Relative humidity	

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13 Appendices

- [Appendix A: 3D-PAWS station inventory coordinates and elevations](#)
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- [Appendix C: Station Costing, fabrication and assembly documentation](#)
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- [Appendix J.1 3D PAWS User Feedback Form & Responses](#)
- [Appendix J.1 3D PAWS User Feedback Form & Responses](#)